

BFS Traversal

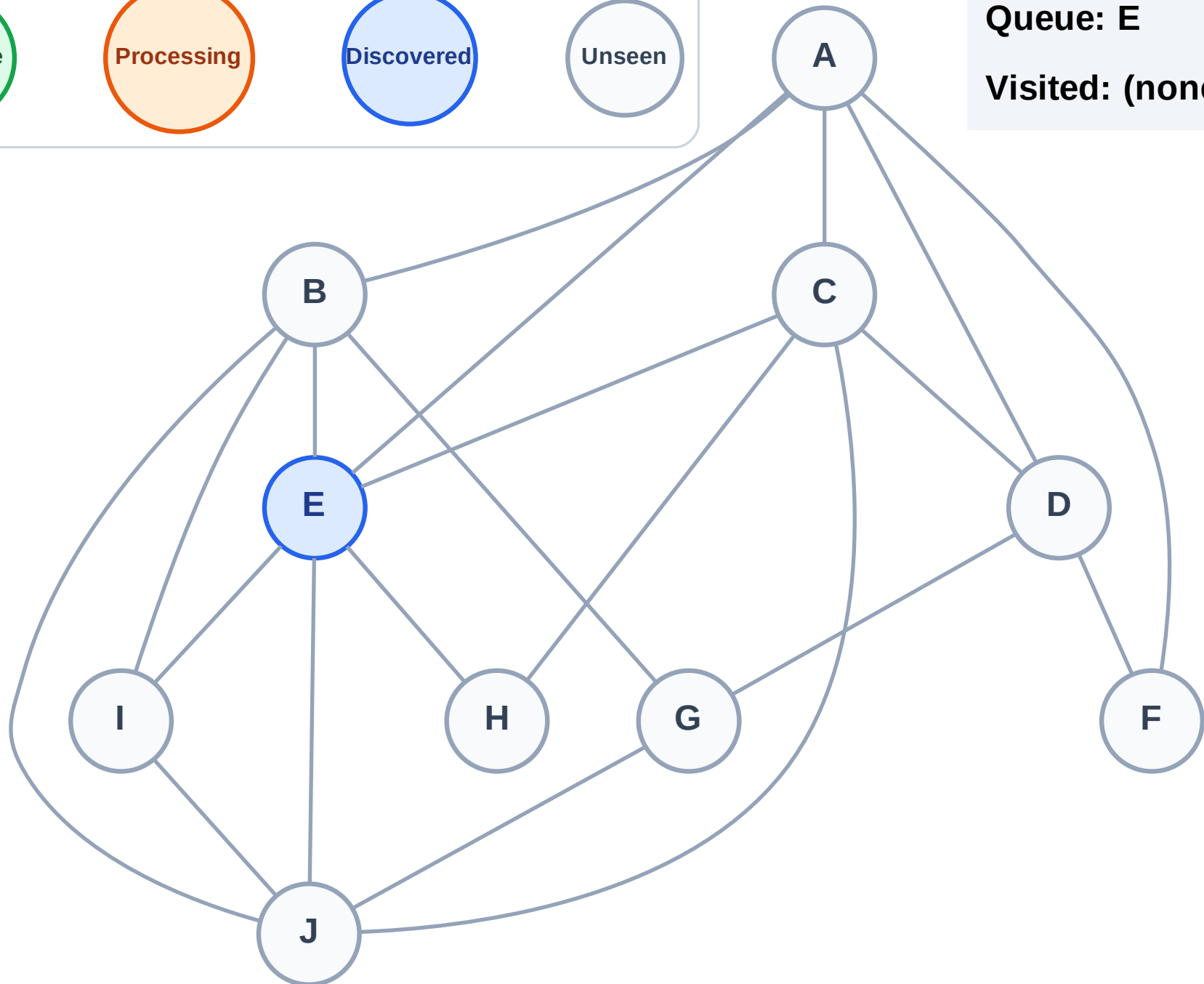
Step 1: Start at E

Legend



Queue: E

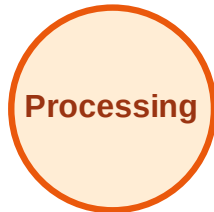
Visited: (none)



BFS Traversal

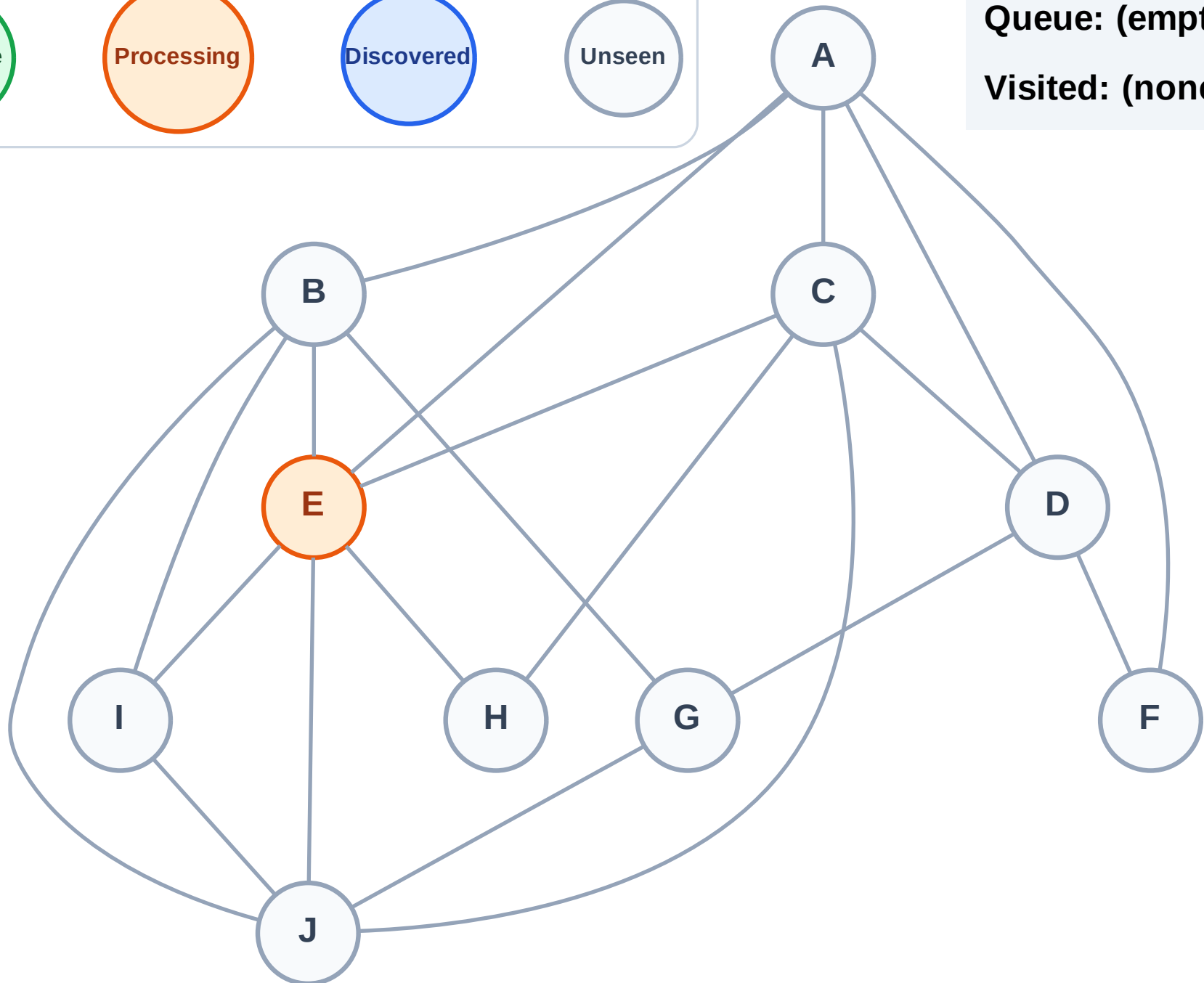
Step 2: Process node E

Legend



Queue: (empty)

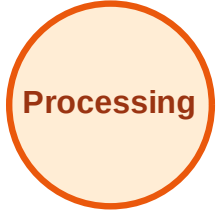
Visited: (none)



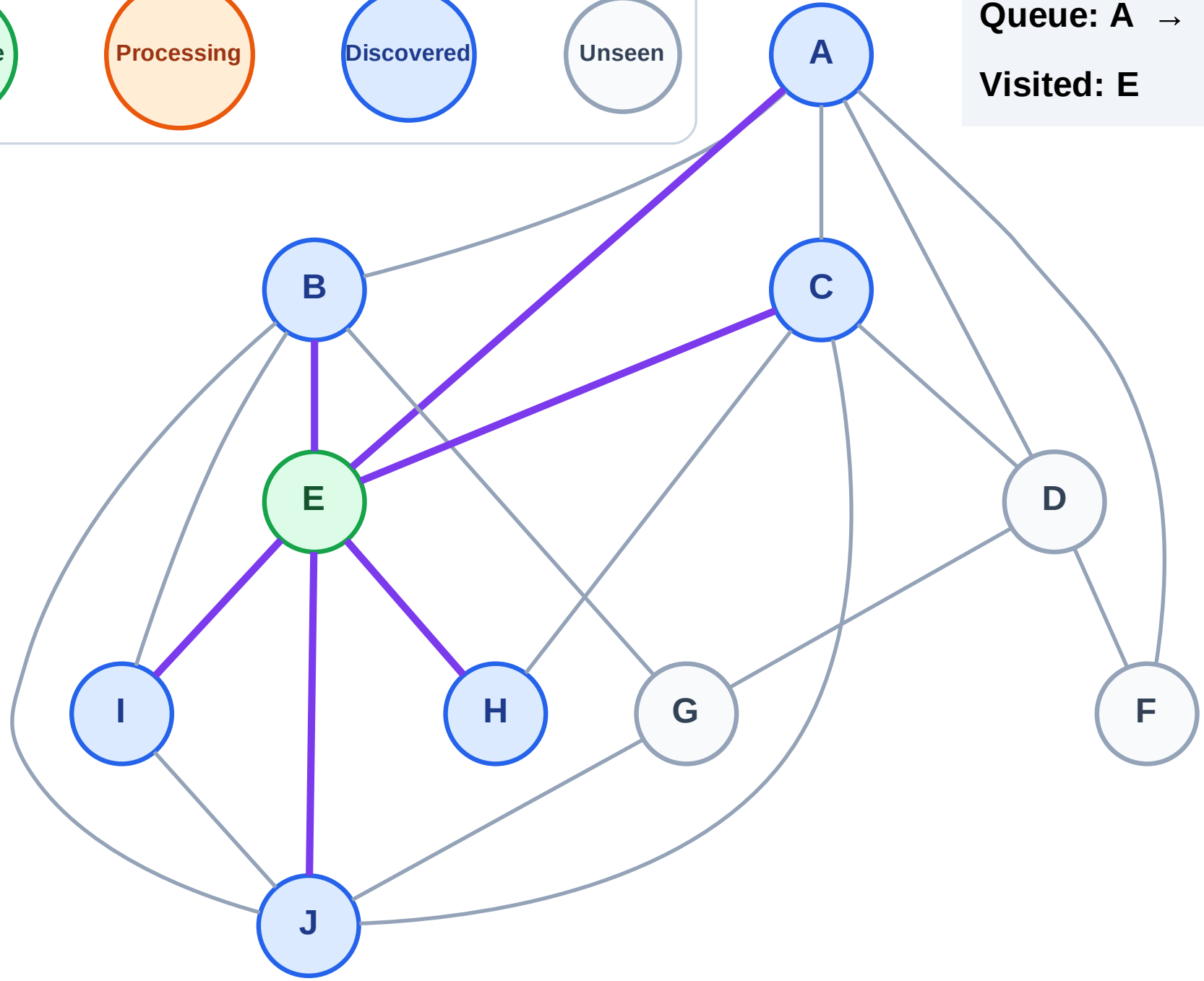
BFS Traversal

Step 3: Discover A, B, C, H, I, J from E

Legend



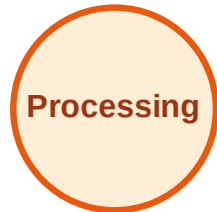
Queue: A → B → C → H → I → J
Visited: E



BFS Traversal

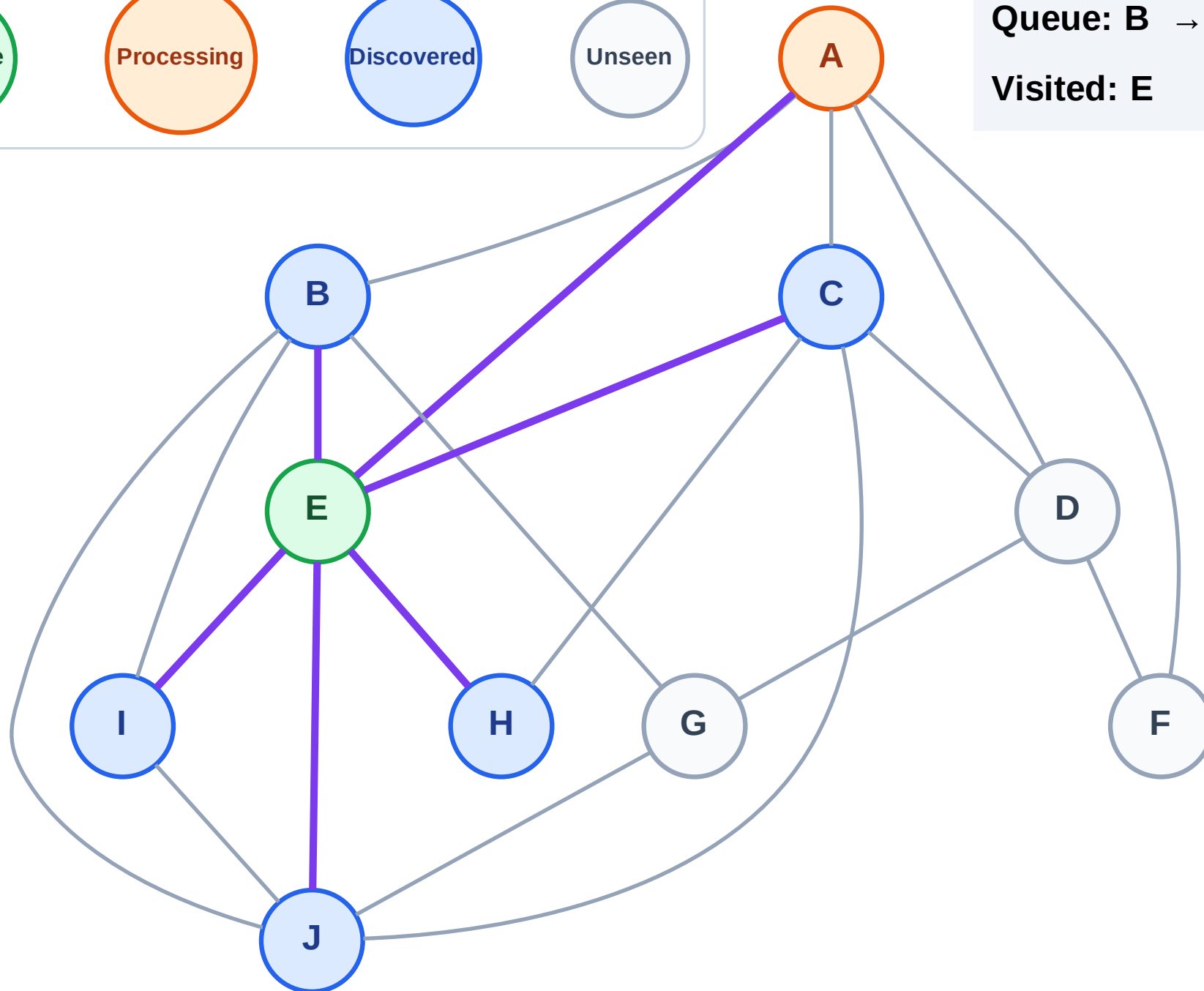
Step 4: Process node A

Legend



Queue: B → C → H → I → J

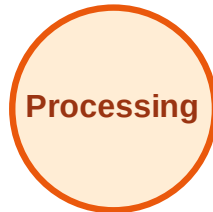
Visited: E



BFS Traversal

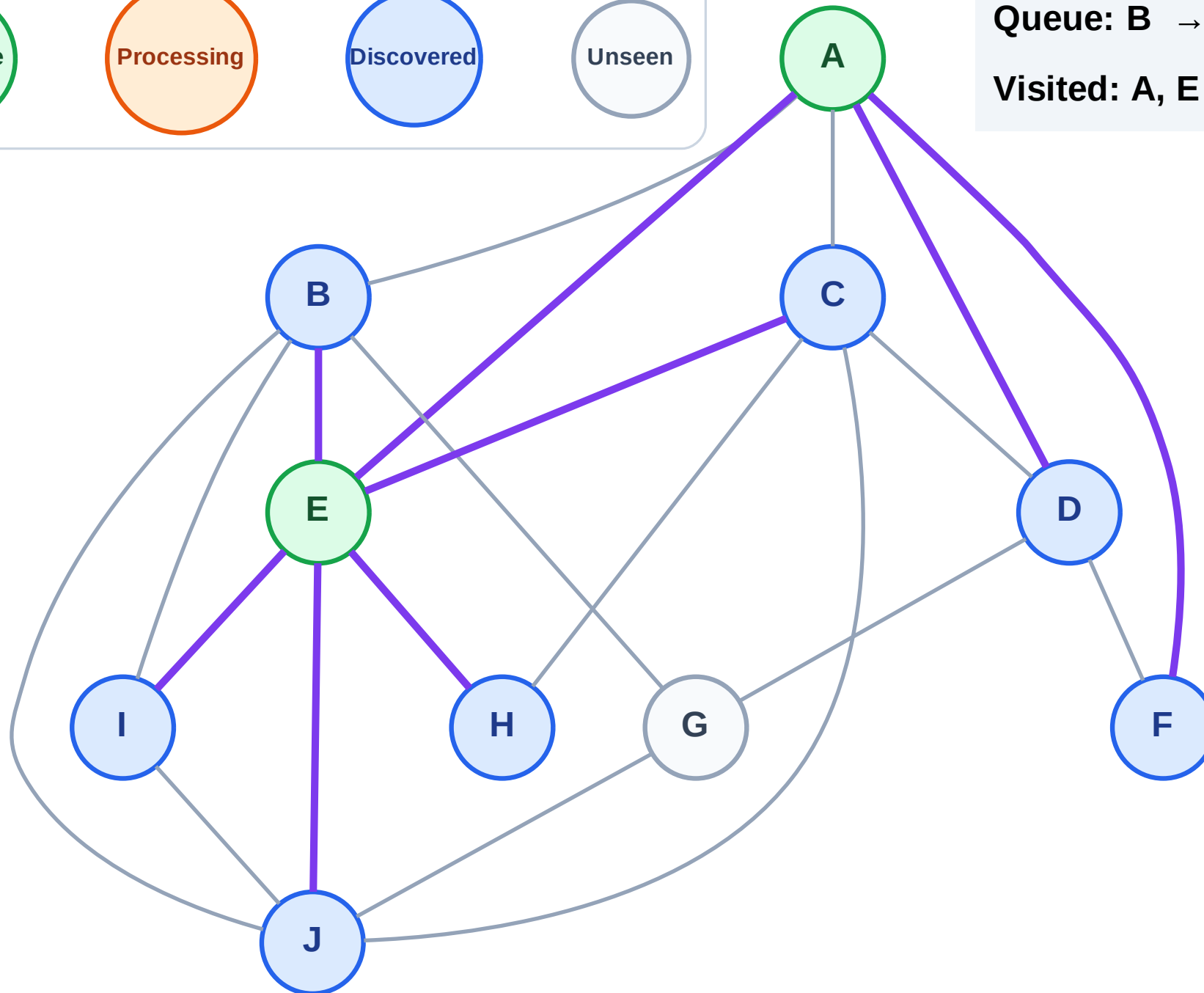
Step 5: Discover D, F from A

Legend



Queue: B → C → H → I → J → D → F

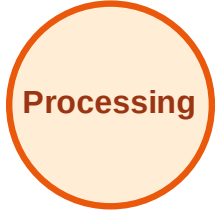
Visited: A, E



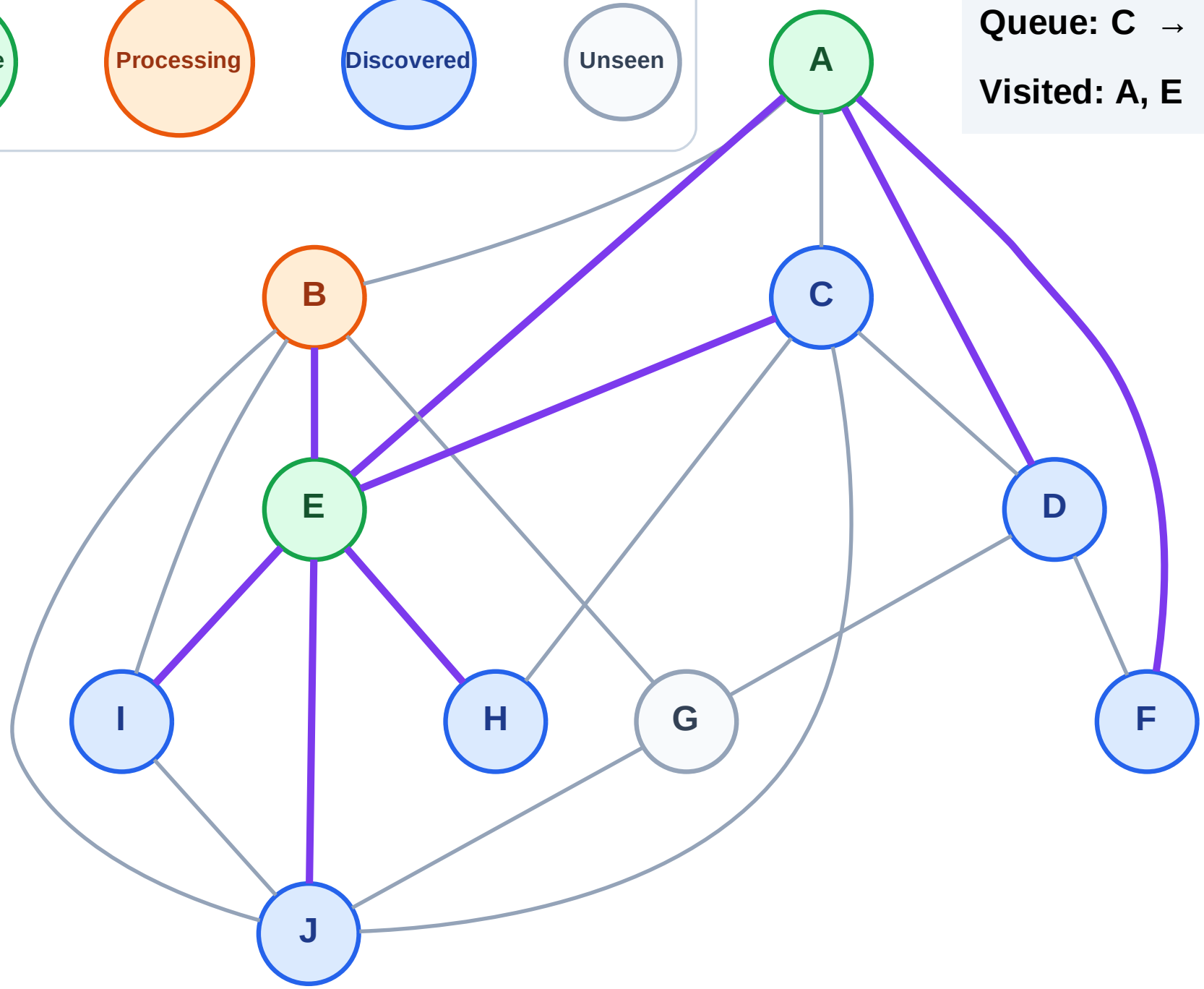
BFS Traversal

Step 6: Process node B

Legend



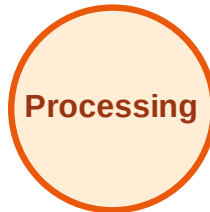
Queue: C → H → I → J → D → F
Visited: A, E



BFS Traversal

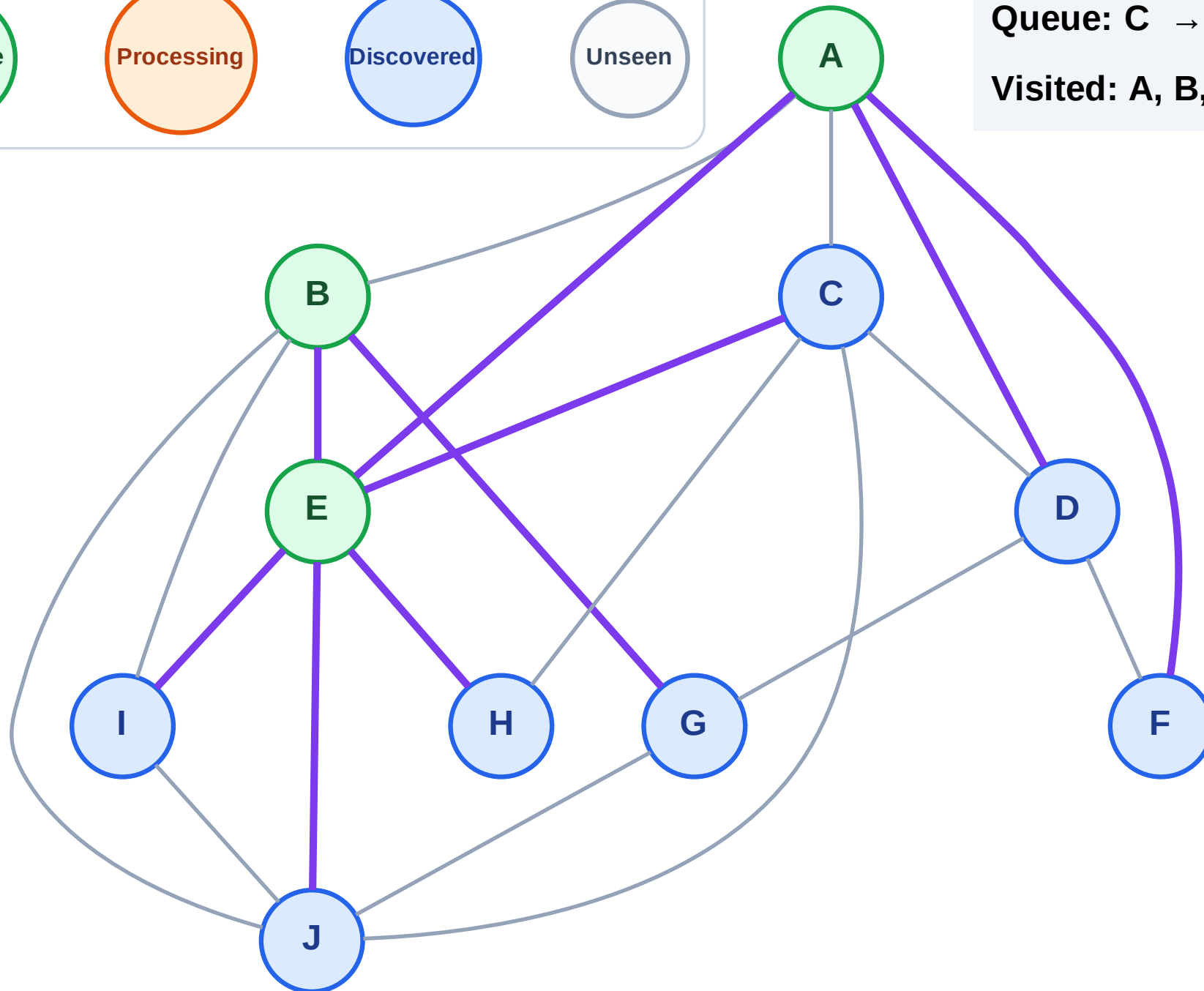
Step 7: Discover G from B

Legend



Queue: C → H → I → J → D → F → G

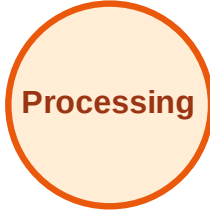
Visited: A, B, E



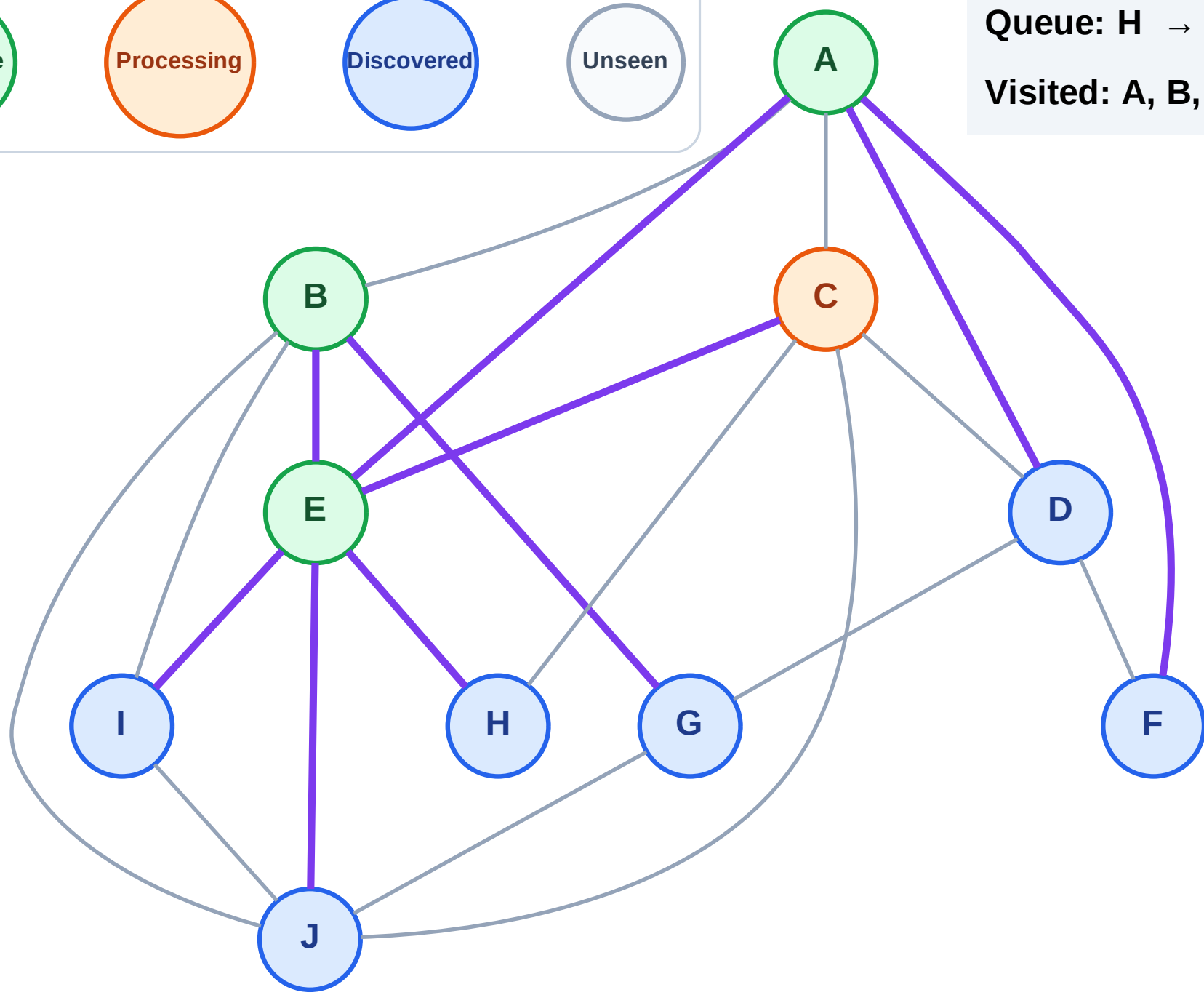
BFS Traversal

Step 8: Process node C

Legend



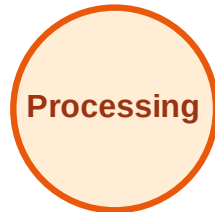
Queue: H → I → J → D → F → G
Visited: A, B, E



BFS Traversal

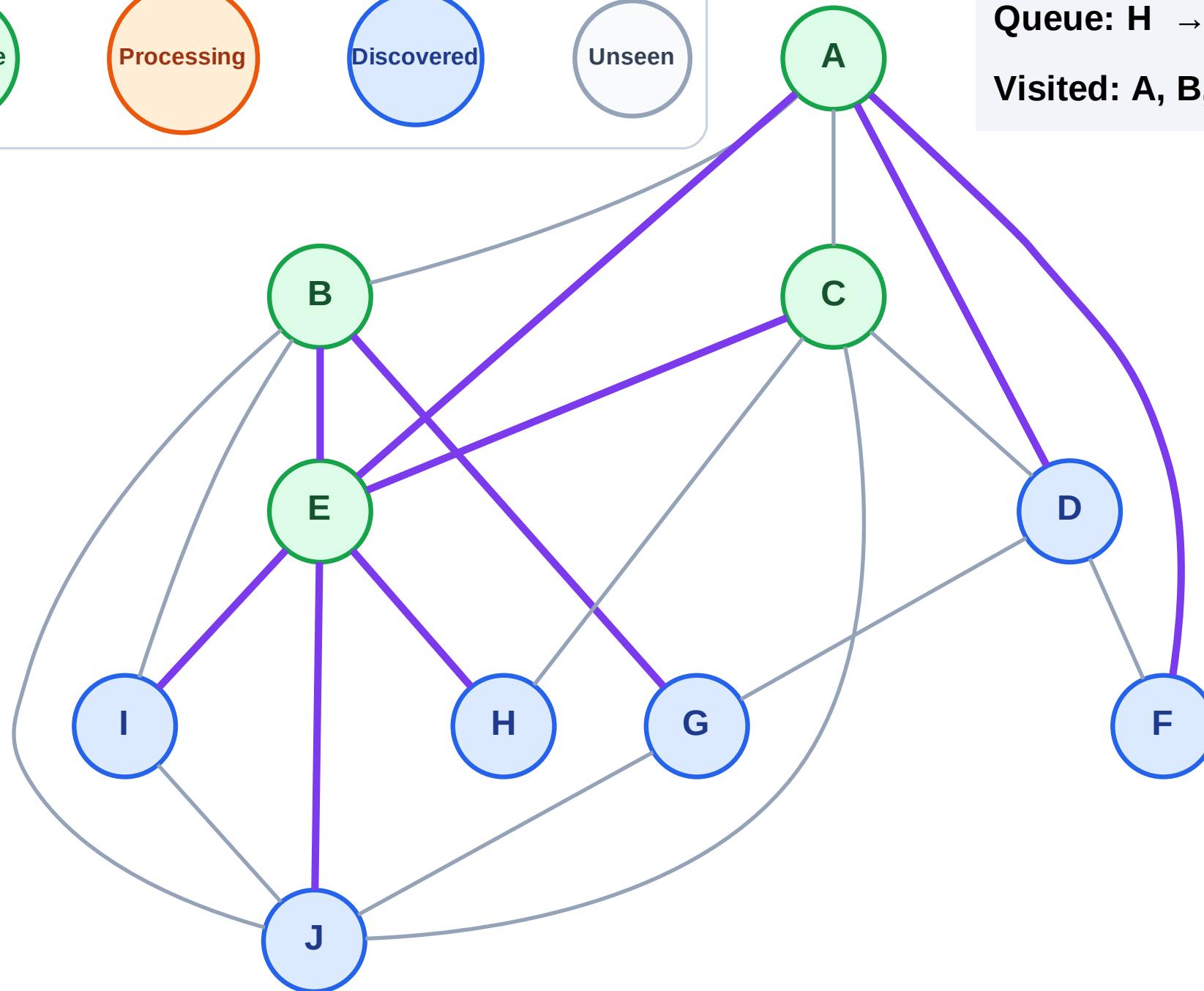
Step 9: No new neighbors from C

Legend



Queue: H → I → J → D → F → G

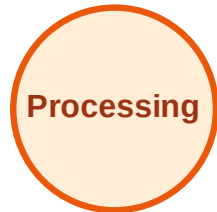
Visited: A, B, C, E



BFS Traversal

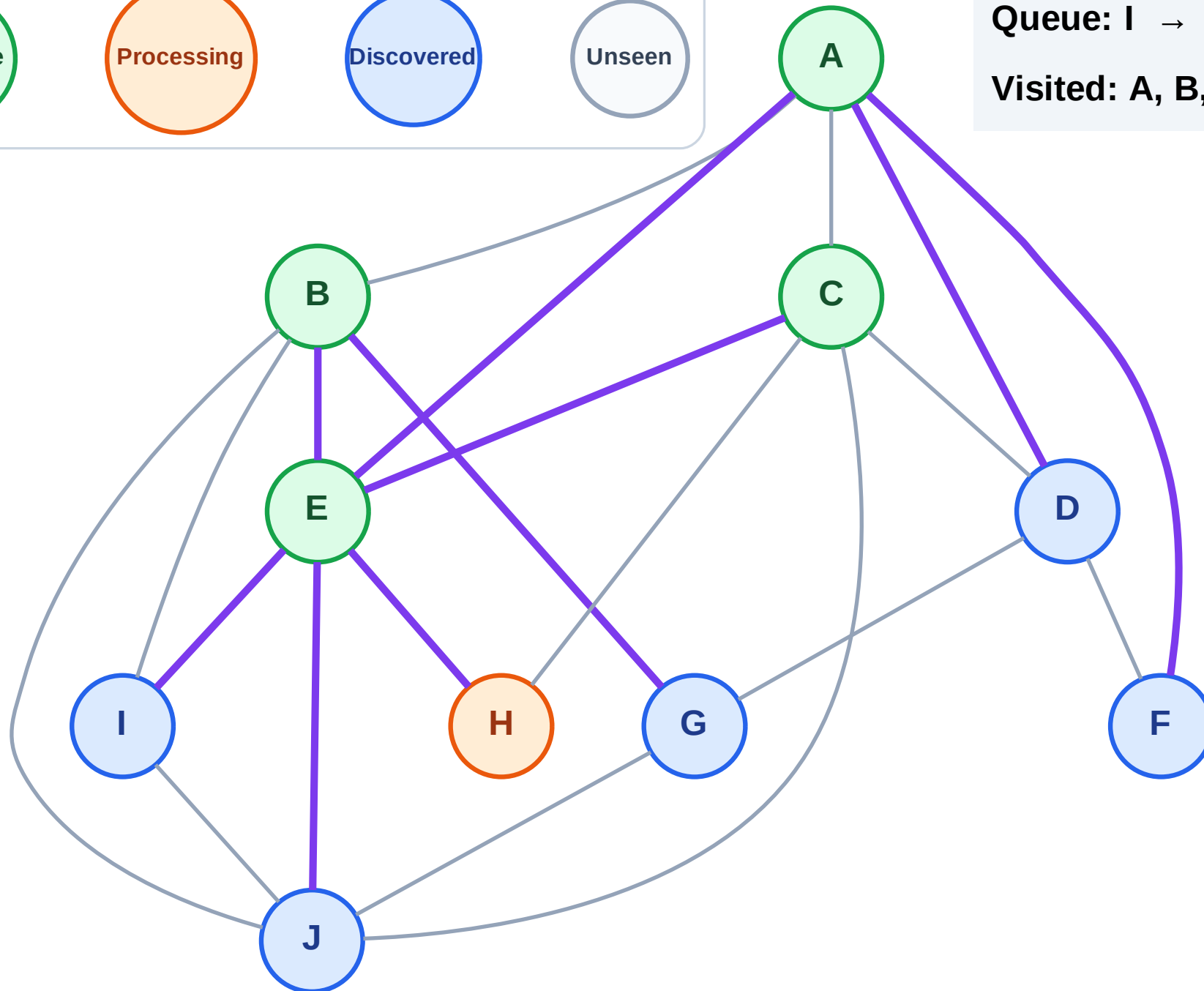
Step 10: Process node H

Legend



Queue: I → J → D → F → G

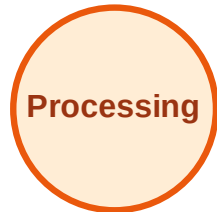
Visited: A, B, C, E



BFS Traversal

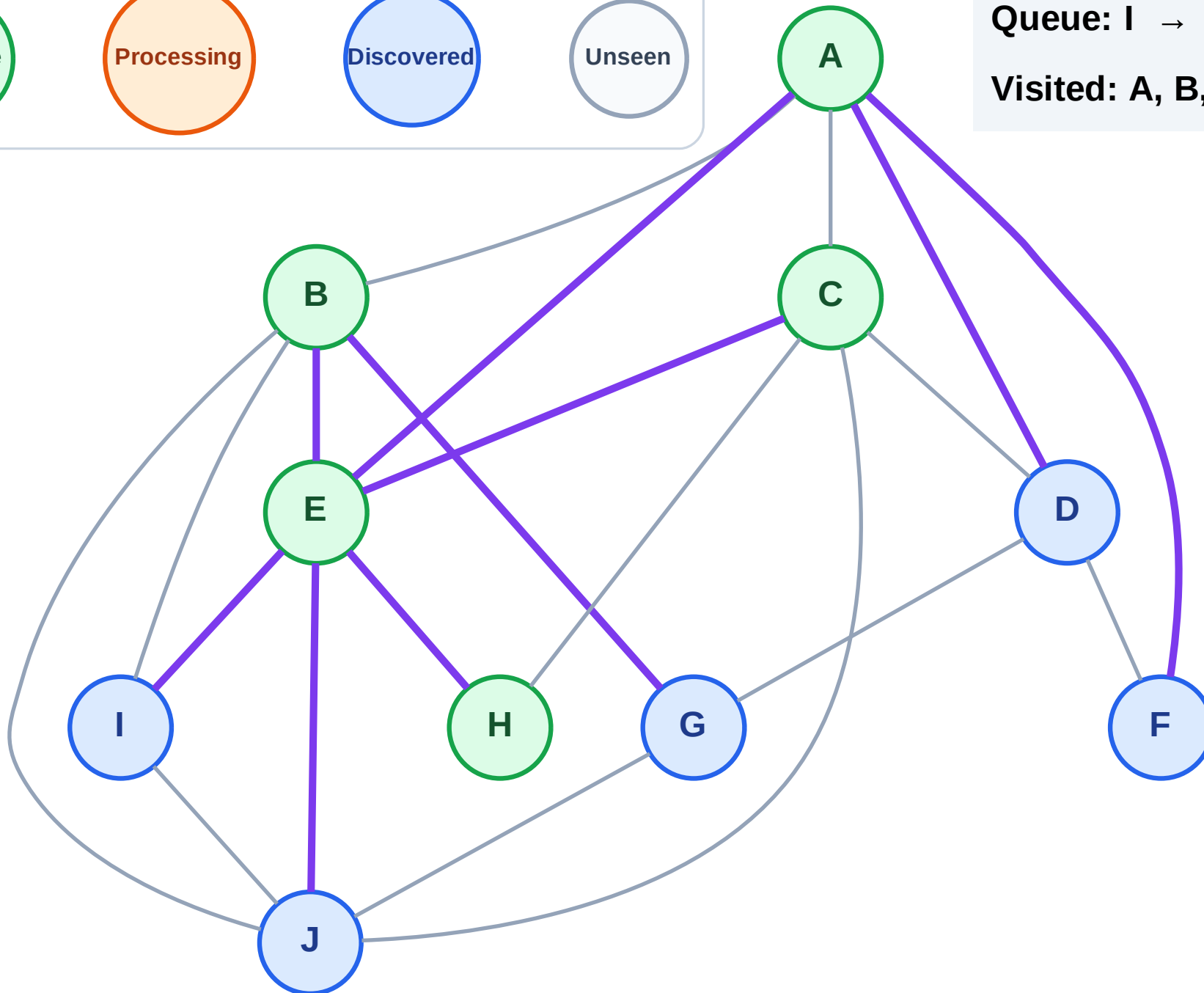
Step 11: No new neighbors from H

Legend



Queue: I → J → D → F → G

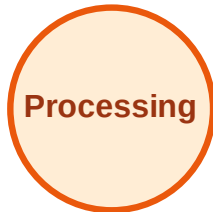
Visited: A, B, C, E, H



BFS Traversal

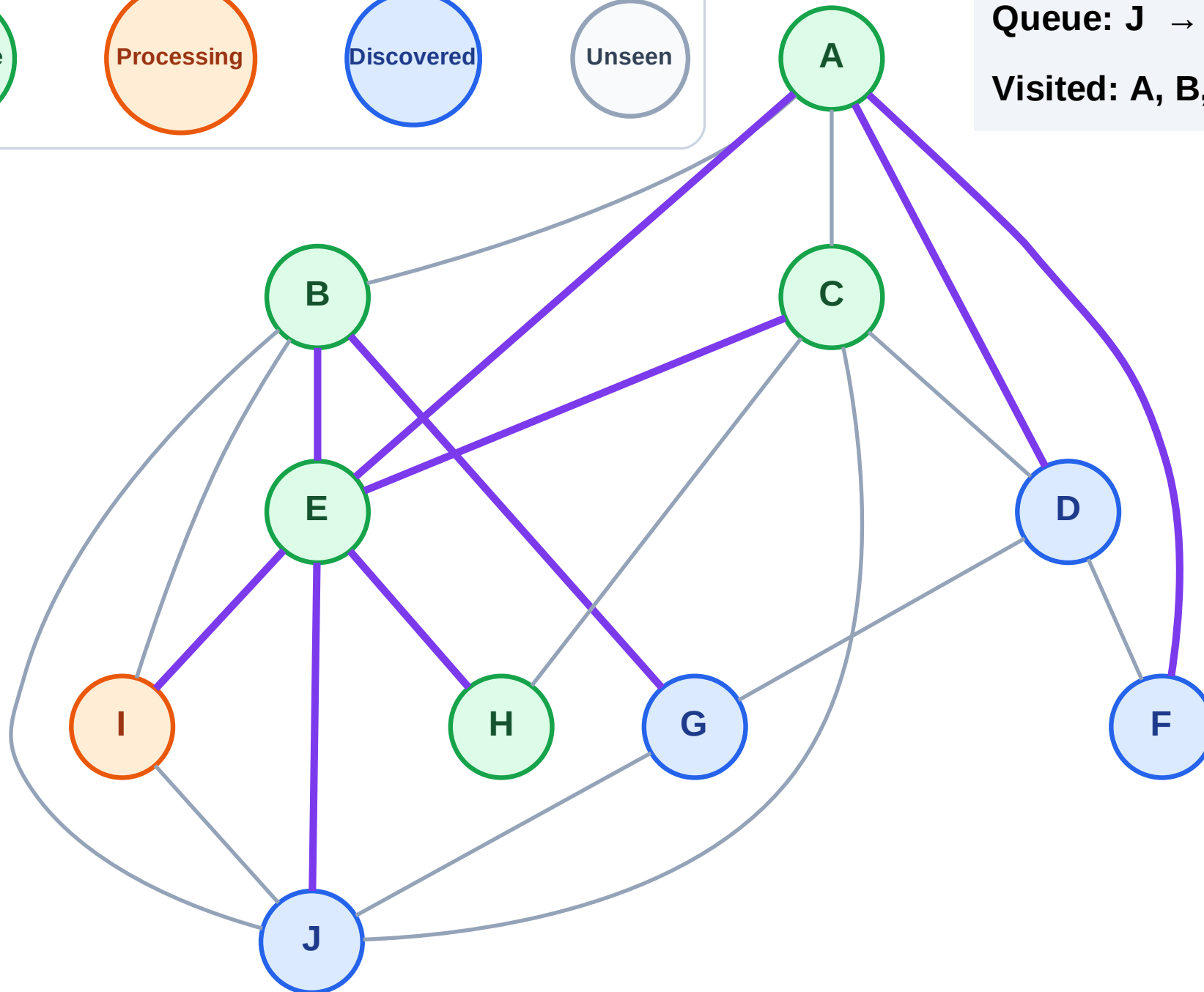
Step 12: Process node I

Legend



Queue: J → D → F → G

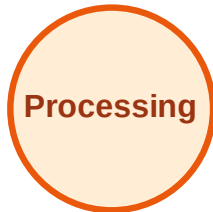
Visited: A, B, C, E, H



BFS Traversal

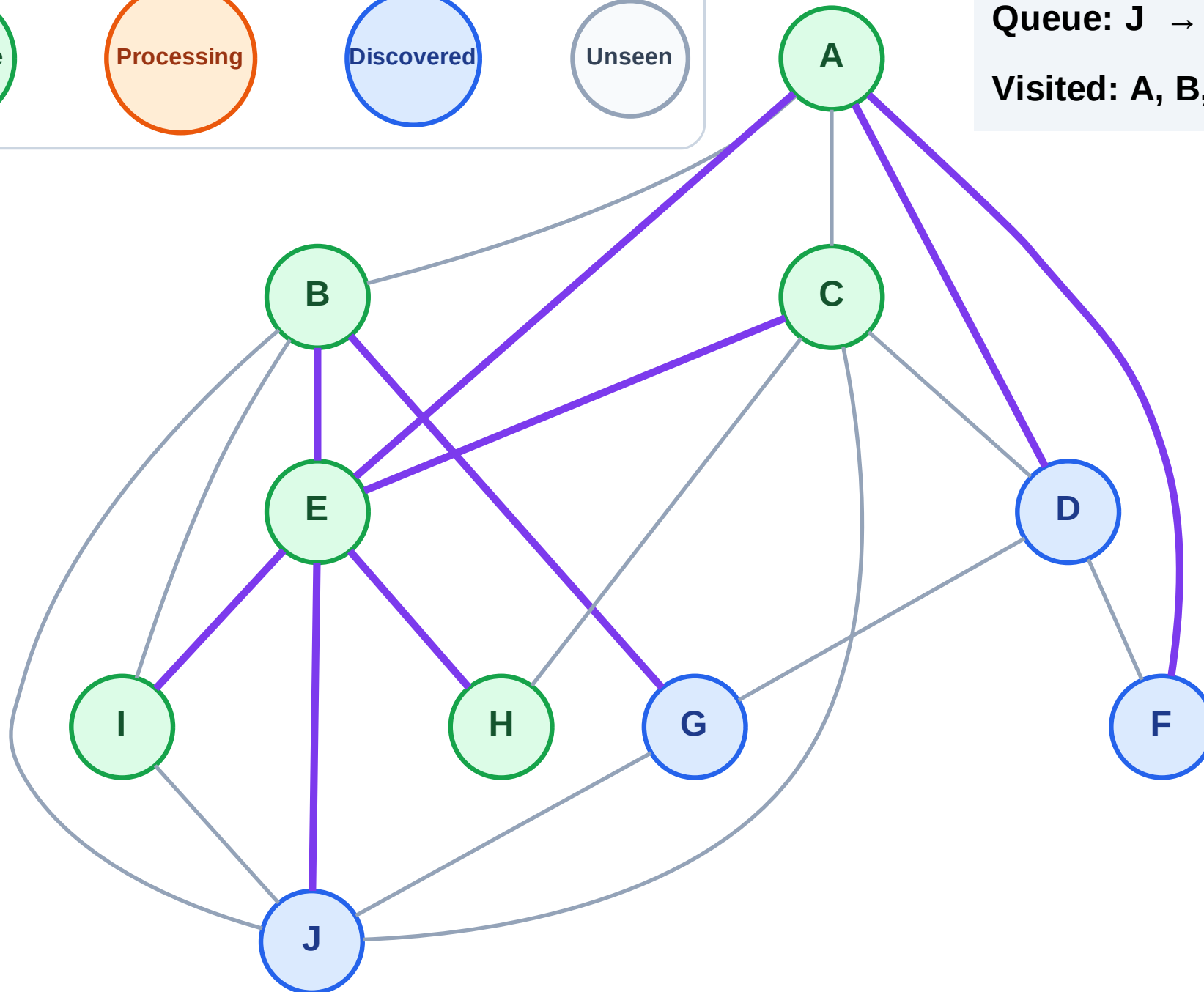
Step 13: No new neighbors from I

Legend



Queue: J → D → F → G

Visited: A, B, C, E, H, I



BFS Traversal

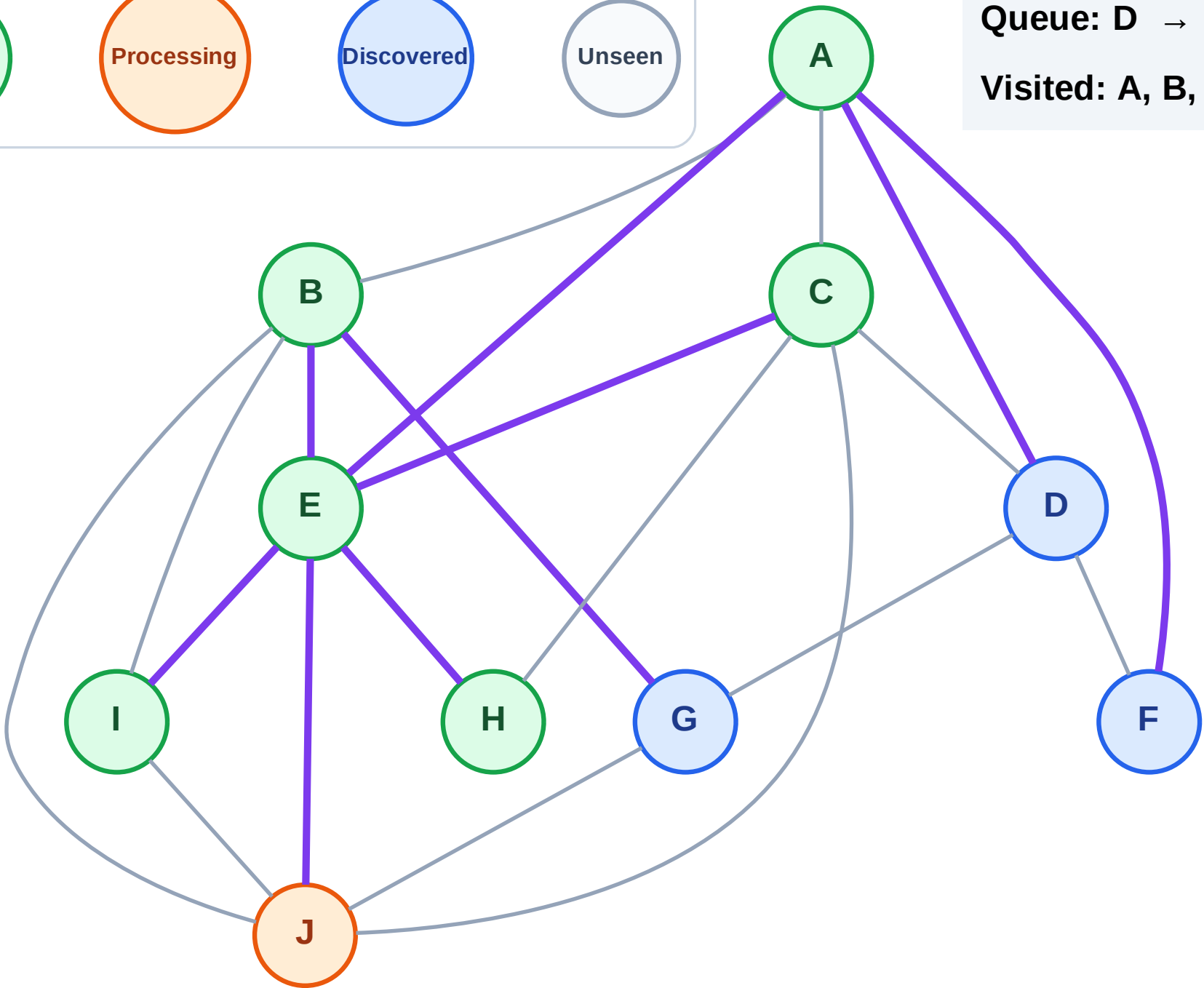
Step 14: Process node J

Legend



Queue: D → F → G

Visited: A, B, C, E, H, I



BFS Traversal

Step 15: No new neighbors from J

Legend

Complete

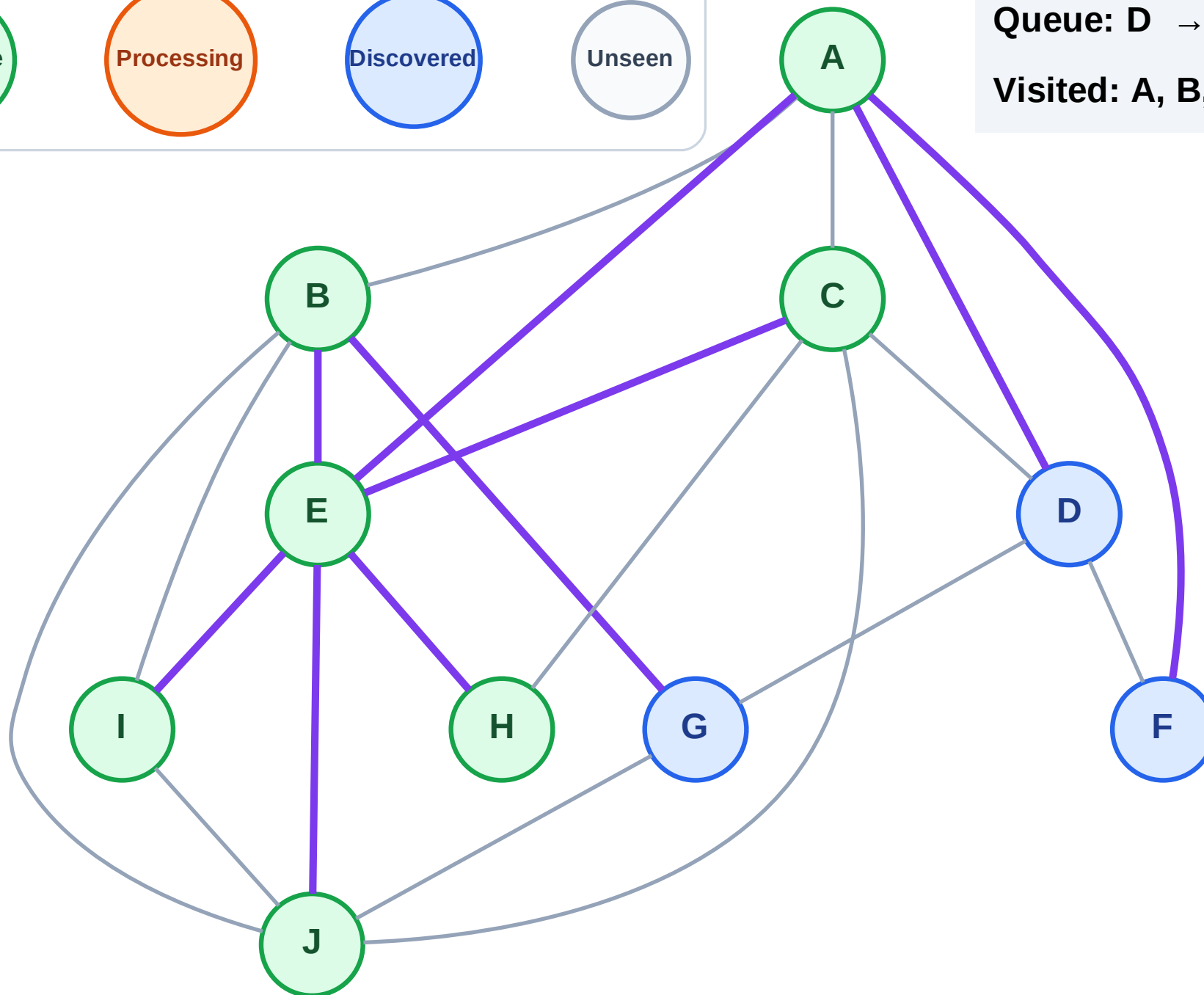
Processing

Discovered

Unseen

Queue: D → F → G

Visited: A, B, C, E, H, I, J



BFS Traversal

Step 16: Process node D

Legend

Complete

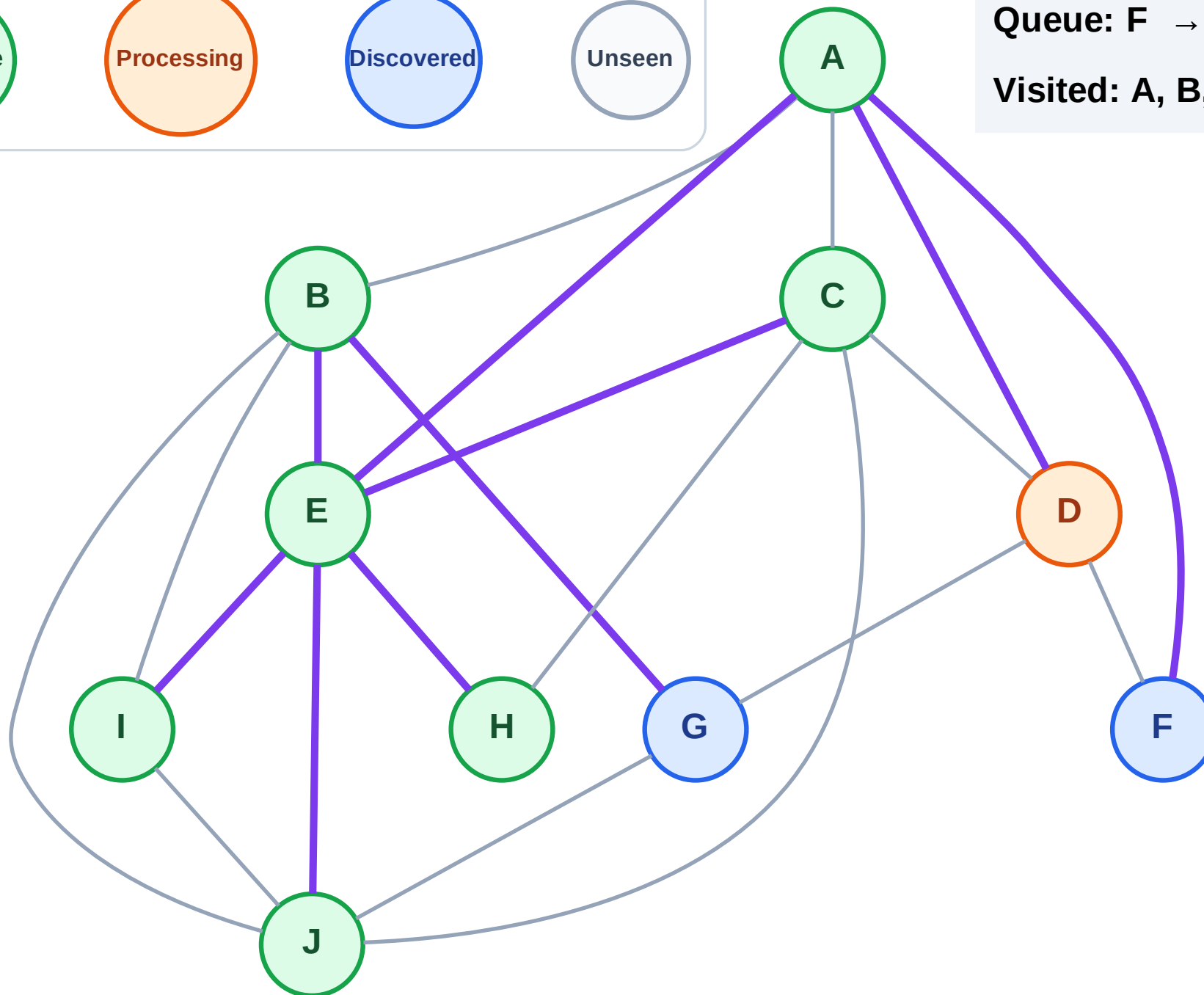
Processing

Discovered

Unseen

Queue: F → G

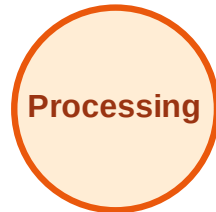
Visited: A, B, C, E, H, I, J



BFS Traversal

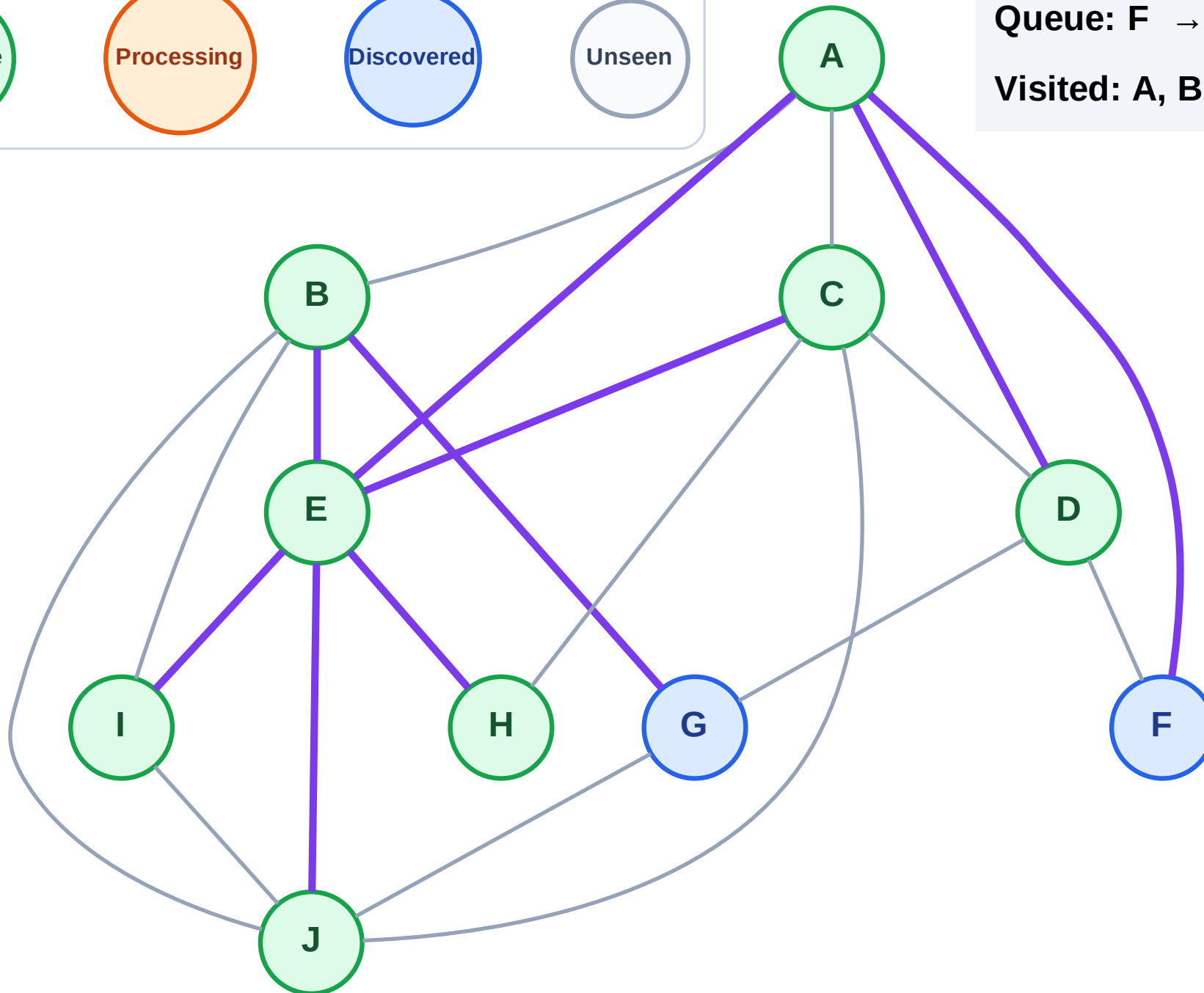
Step 17: No new neighbors from D

Legend



Queue: F → G

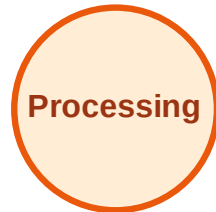
Visited: A, B, C, D, E, H, I, J



BFS Traversal

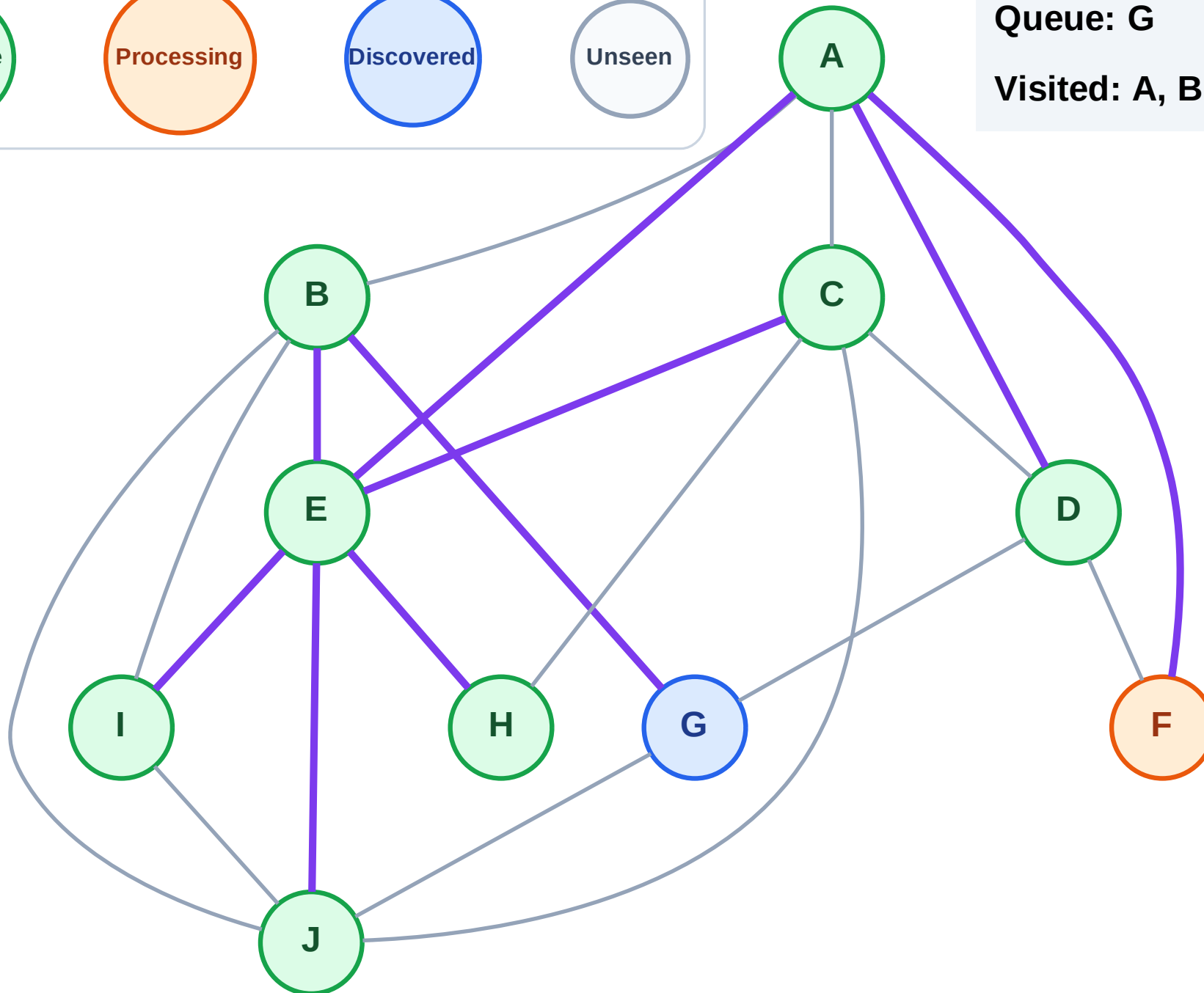
Step 18: Process node F

Legend



Queue: G

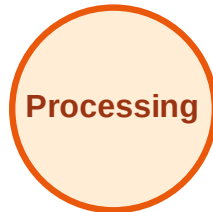
Visited: A, B, C, D, E, H, I, J



BFS Traversal

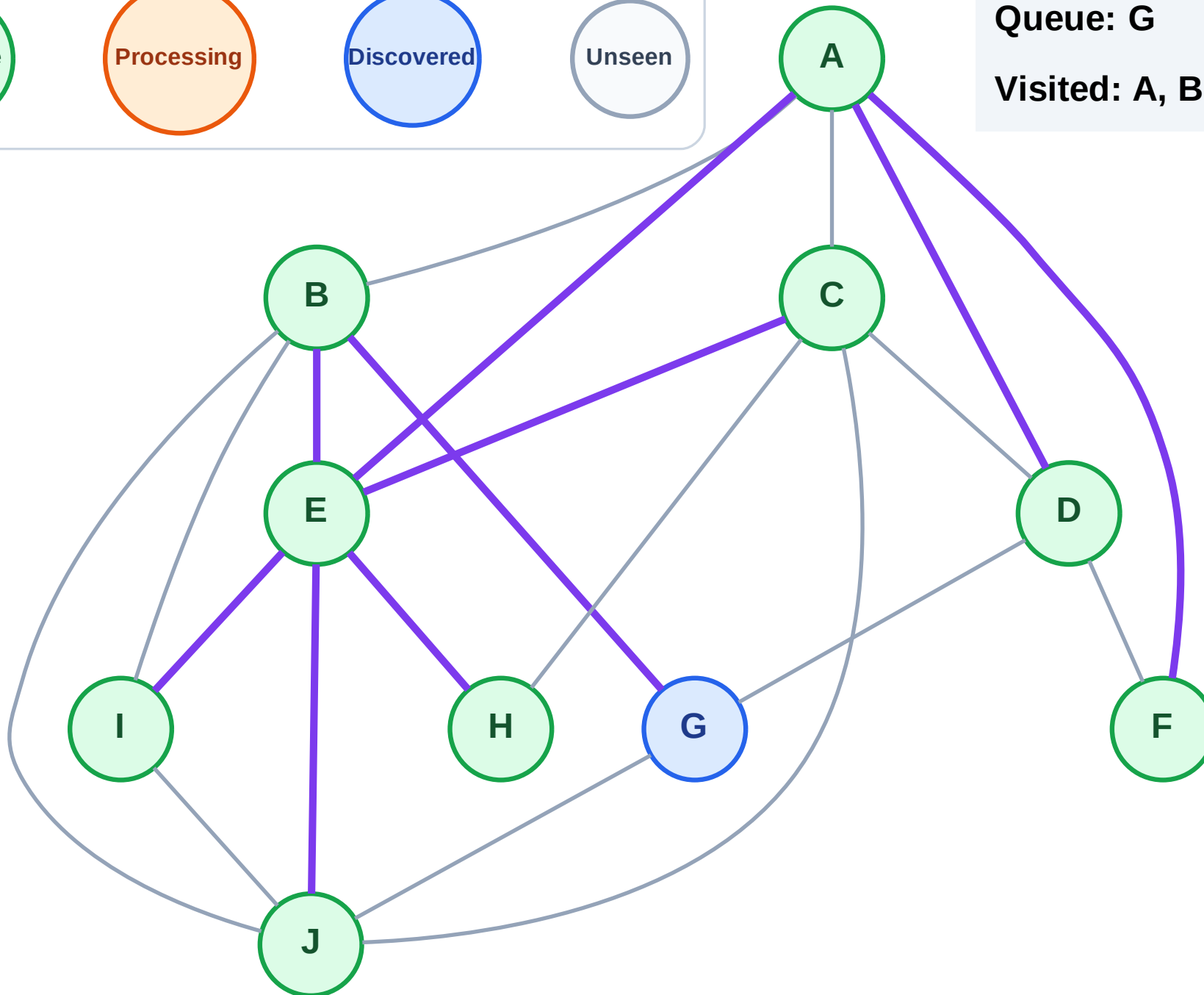
Step 19: No new neighbors from F

Legend



Queue: G

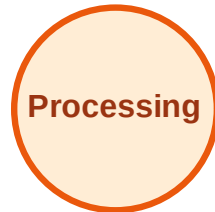
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

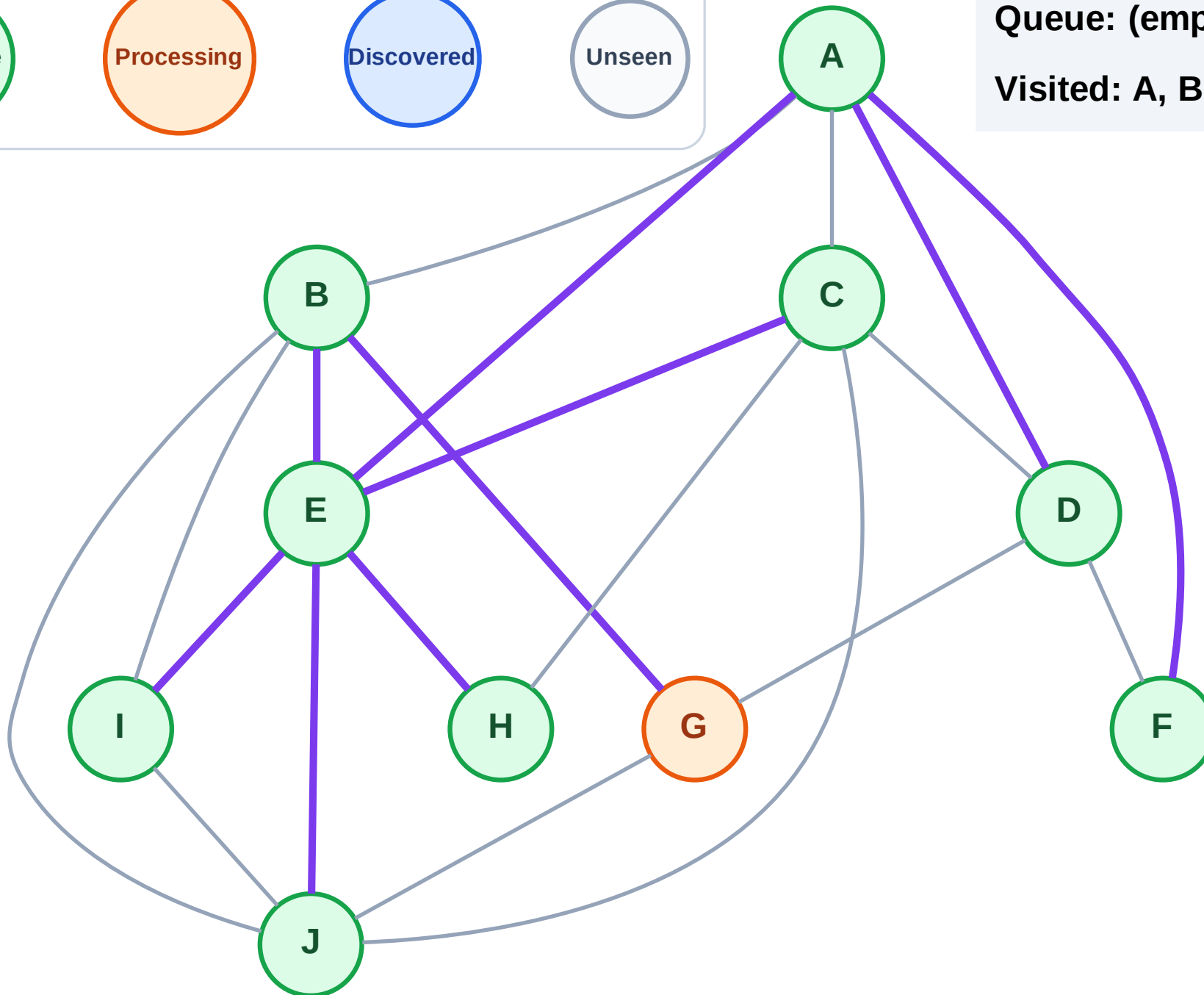
Step 20: Process node G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

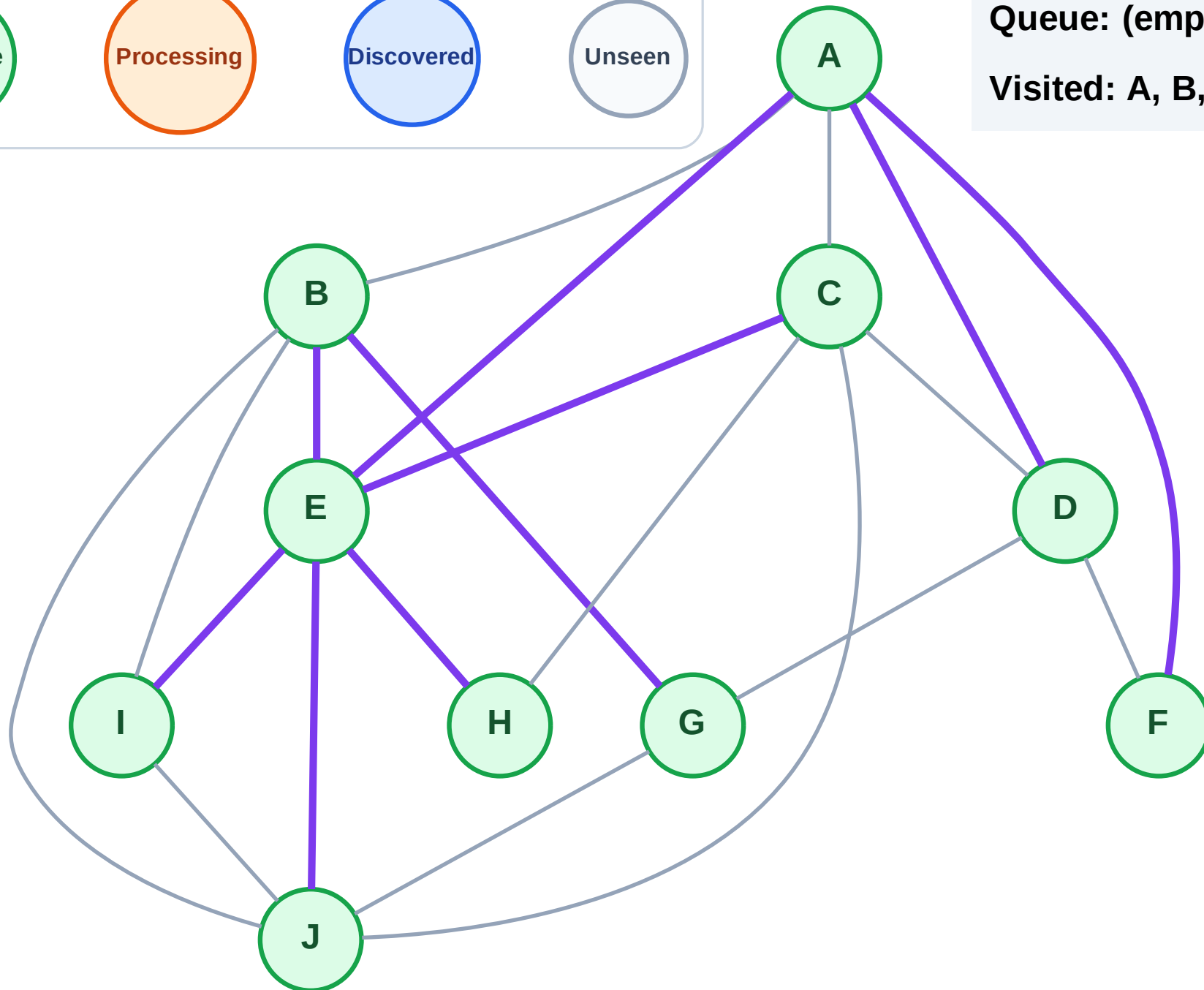
Step 21: No new neighbors from G

Legend



Queue: (empty)

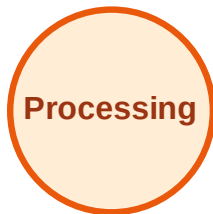
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

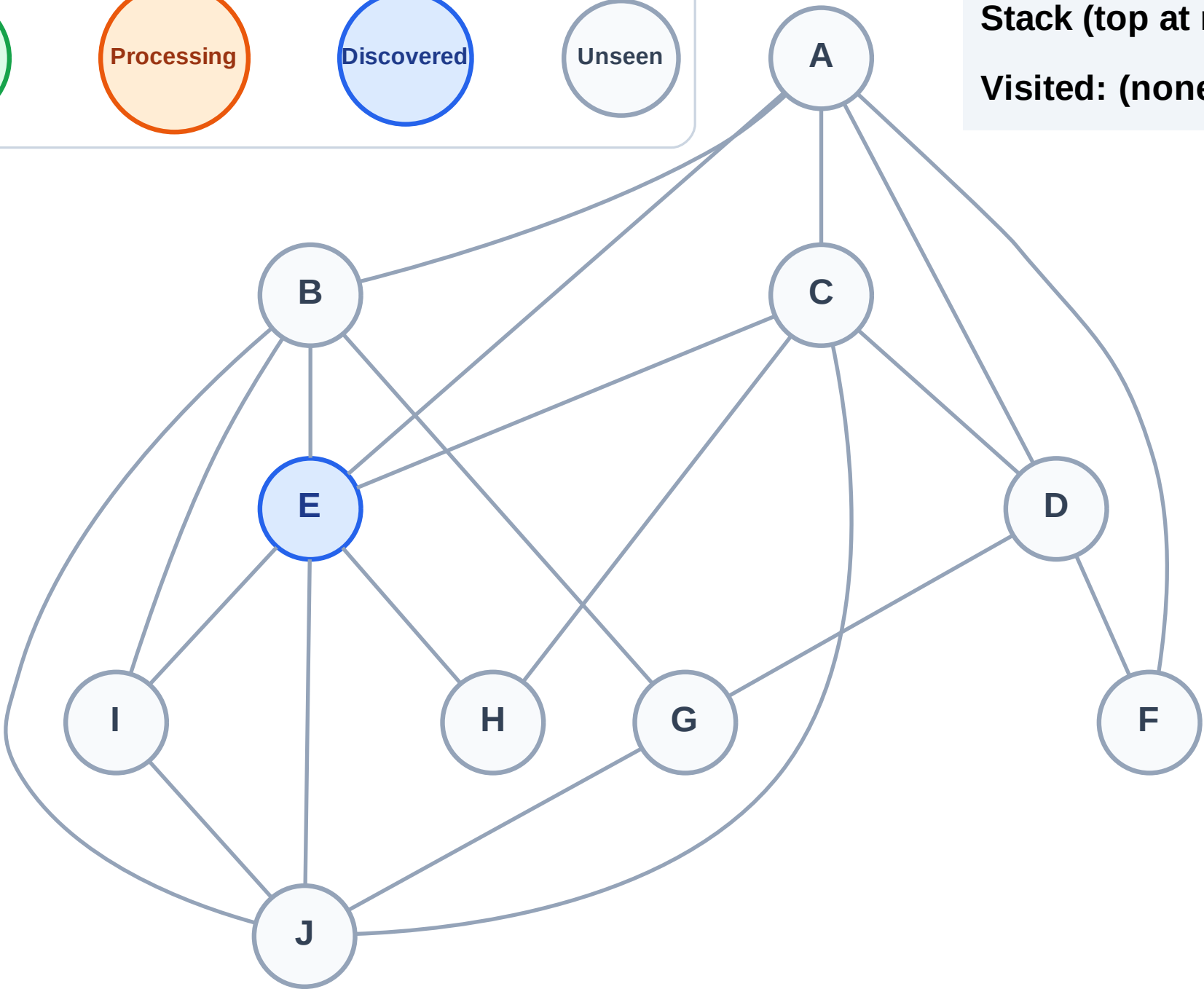
Step 1: Start at E

Legend



Stack (top at right): E

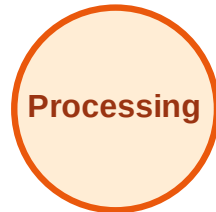
Visited: (none)



DFS Traversal

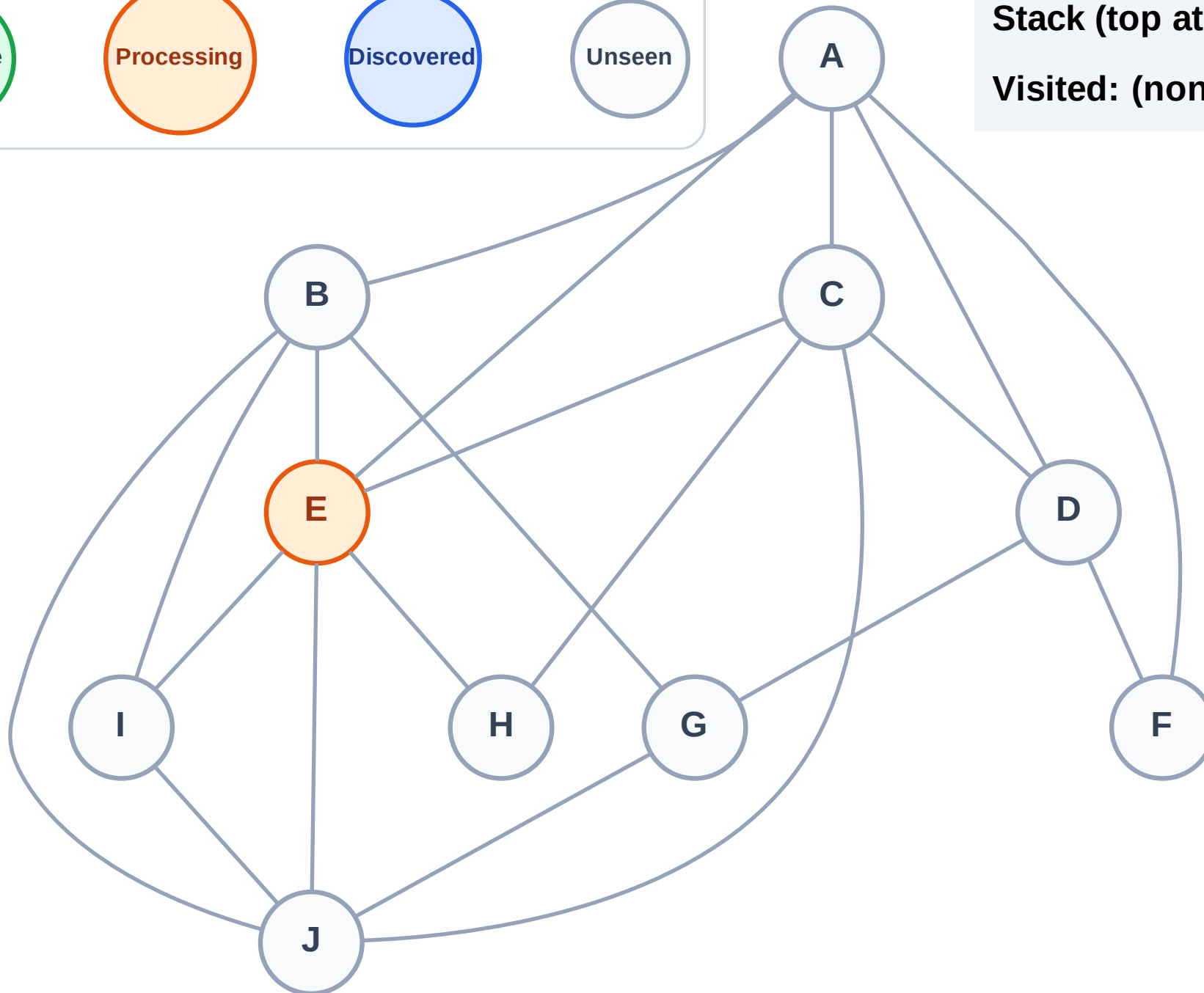
Step 2: Process node E

Legend



Stack (top at right): (empty)

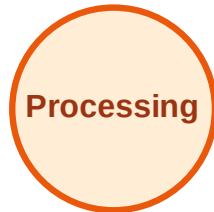
Visited: (none)



DFS Traversal

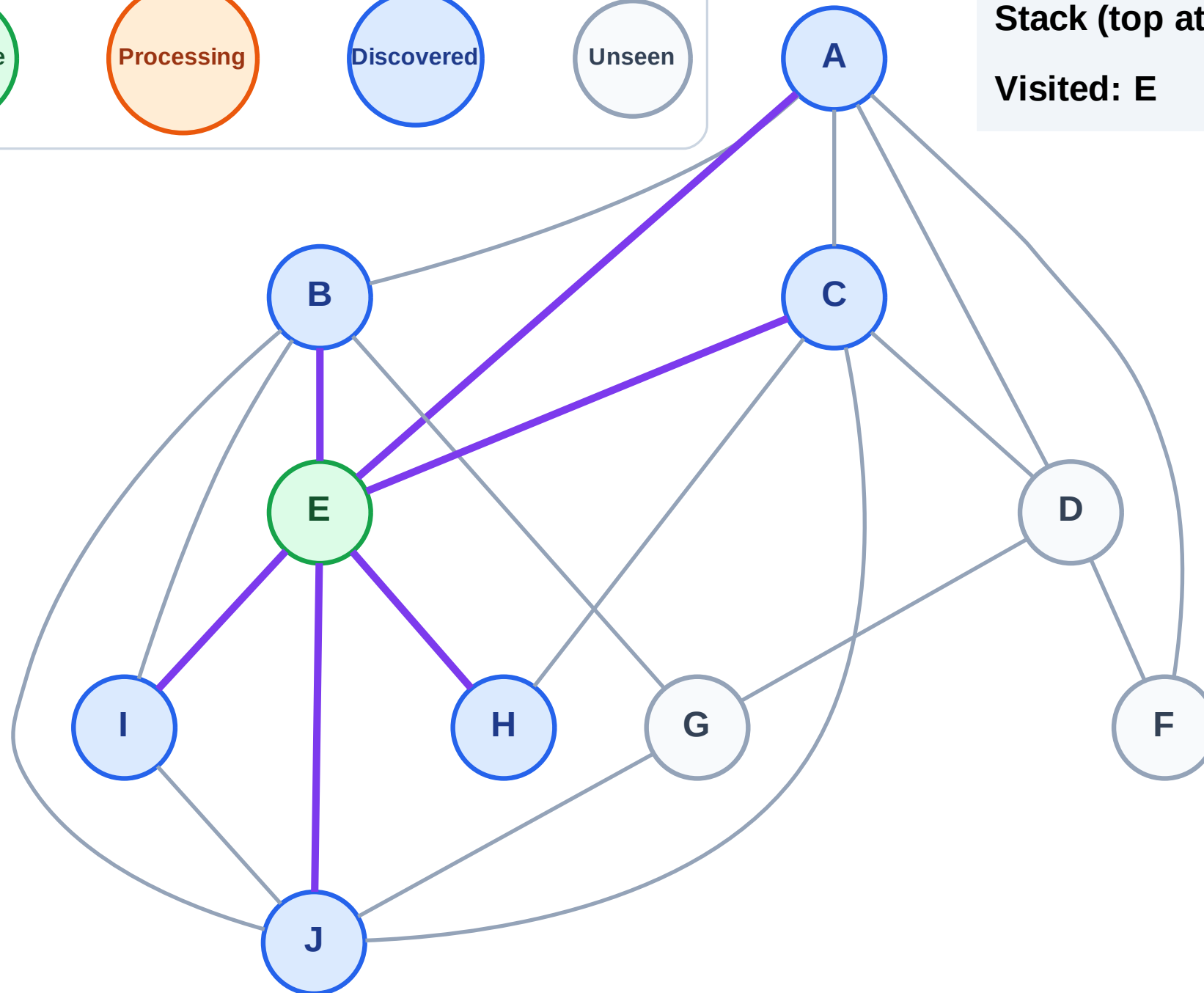
Step 3: Discover J, I, H, C, B, A from E

Legend



Stack (top at right): J | I | H | C | B | A

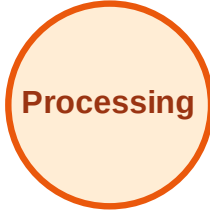
Visited: E



DFS Traversal

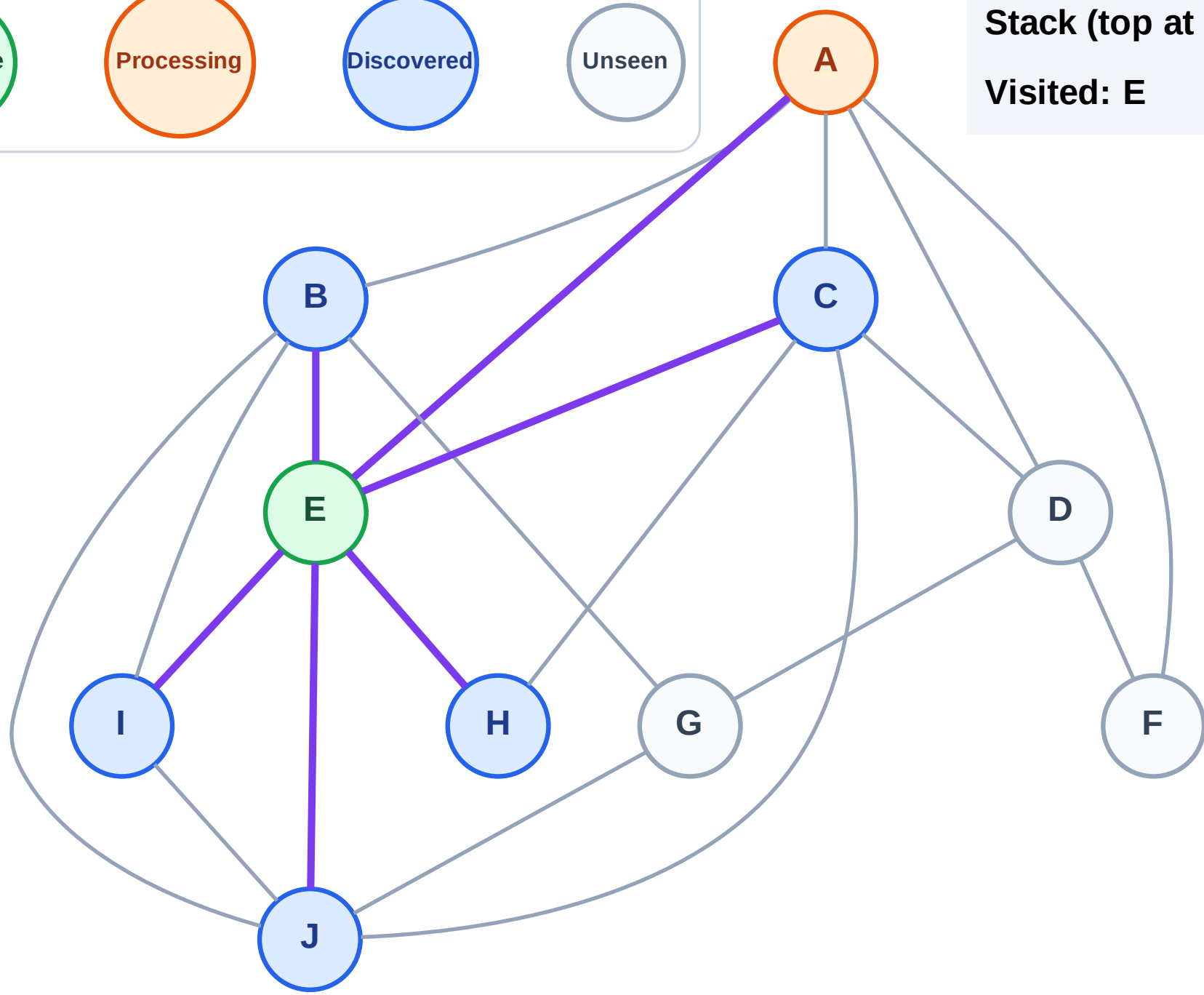
Step 4: Process node A

Legend



Stack (top at right): J | I | H | C | B

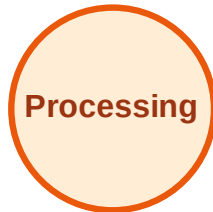
Visited: E



DFS Traversal

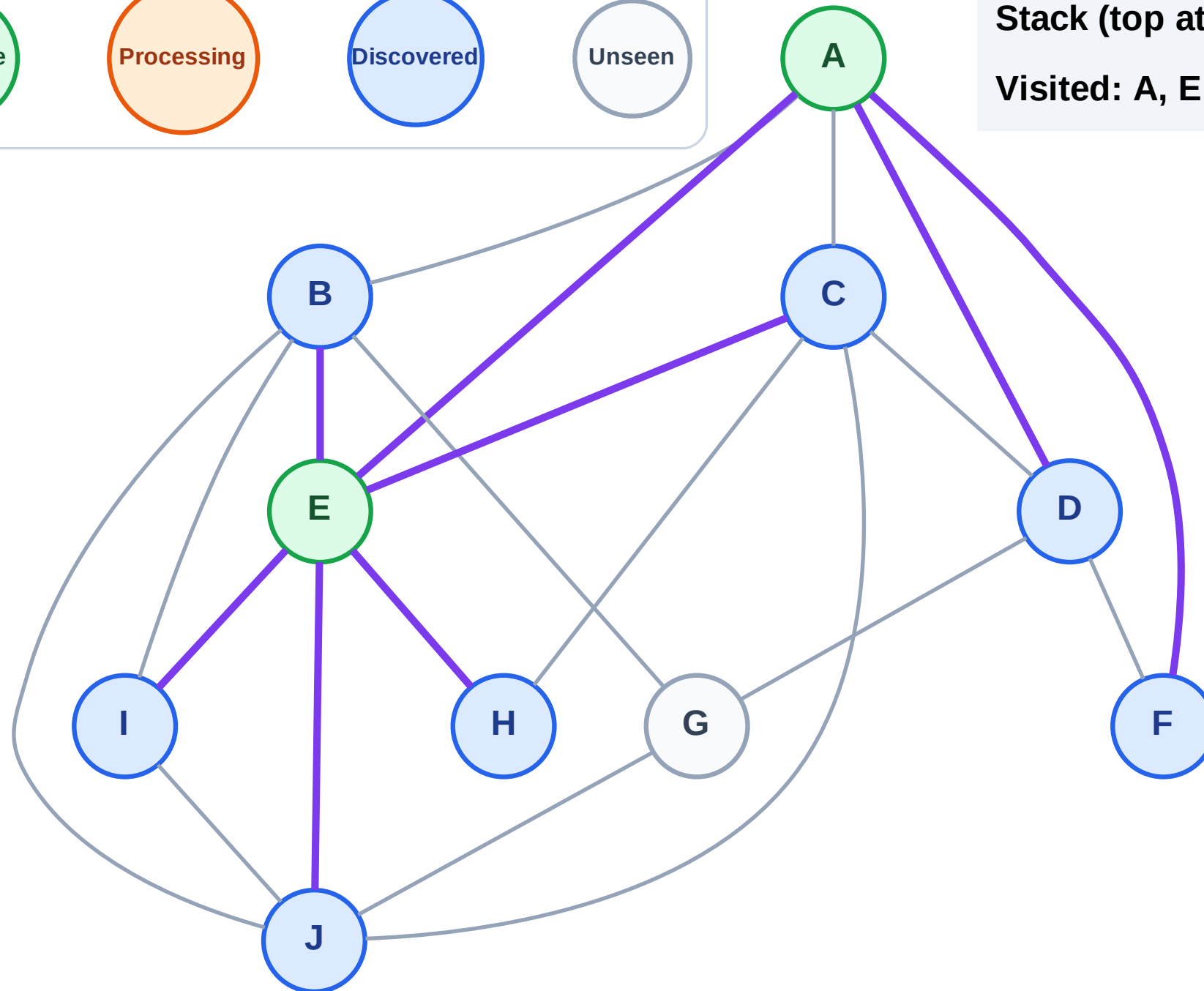
Step 5: Discover F, D from A

Legend



Stack (top at right): J | I | H | C | B | F | D

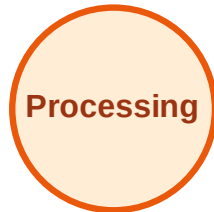
Visited: A, E



DFS Traversal

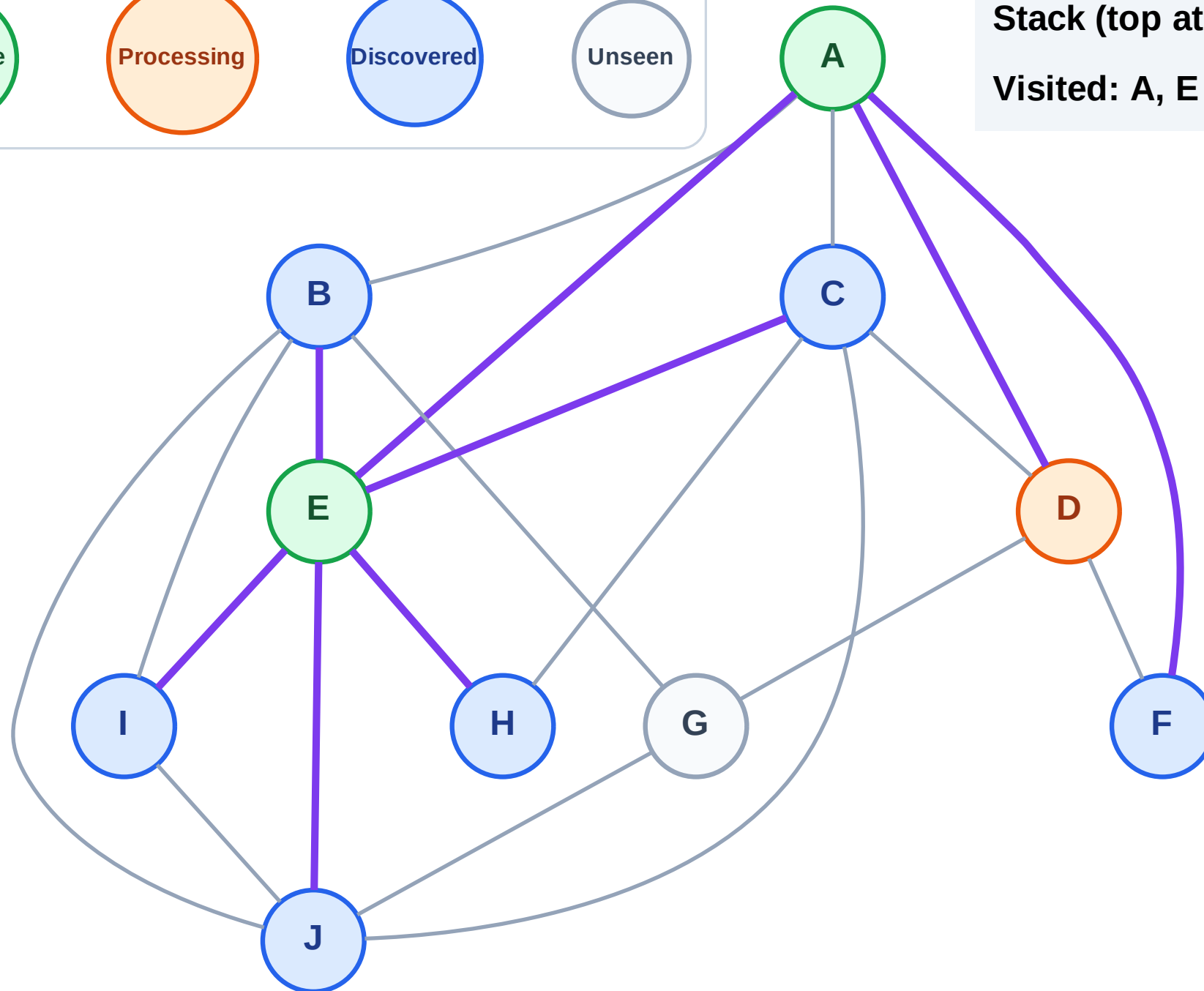
Step 6: Process node D

Legend



Stack (top at right): J | I | H | C | B | F

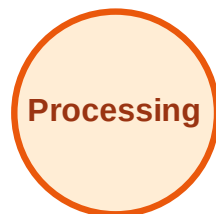
Visited: A, E



DFS Traversal

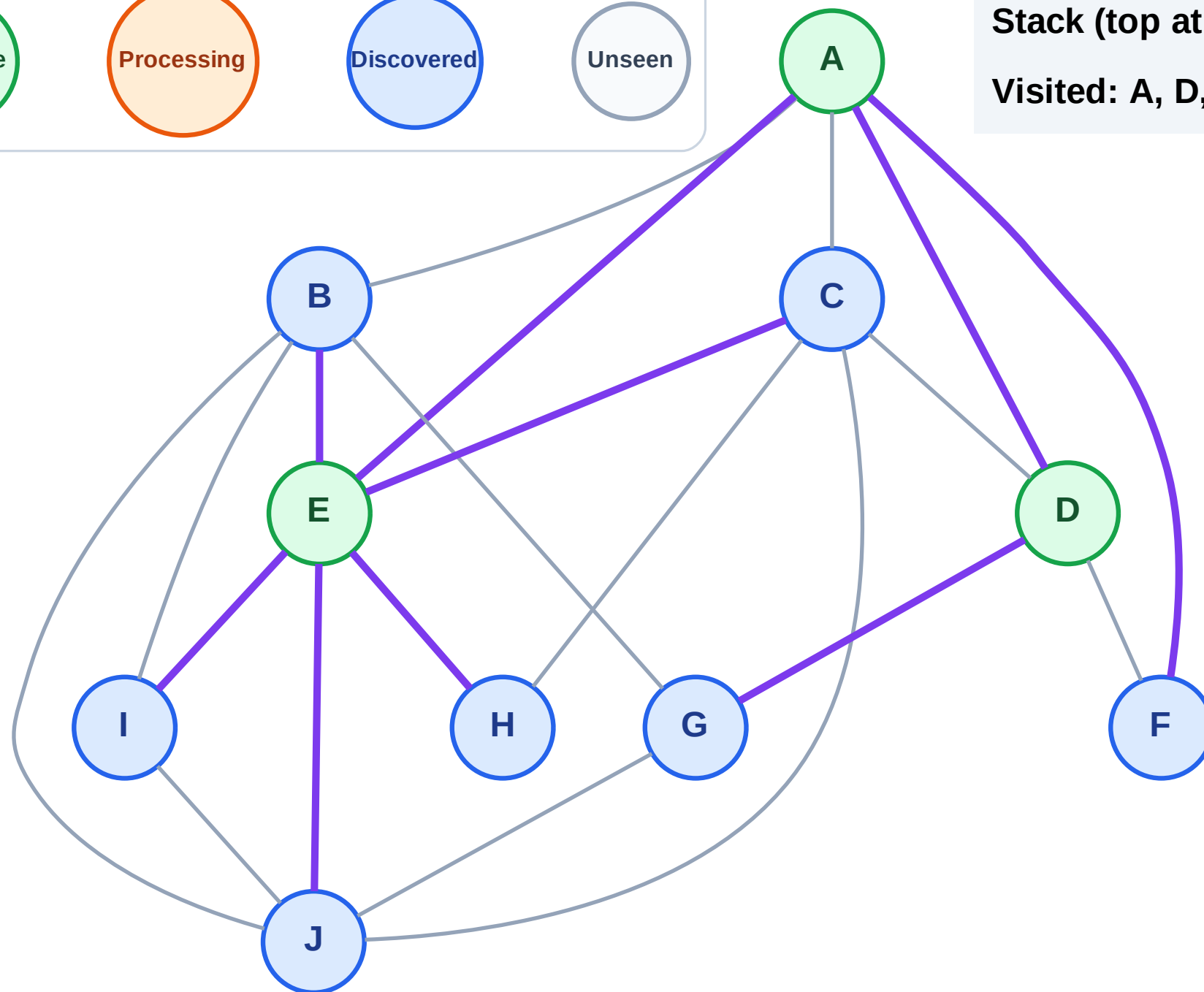
Step 7: Discover G from D

Legend



Stack (top at right): J | I | H | C | B | F | G

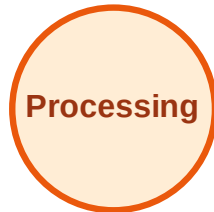
Visited: A, D, E



DFS Traversal

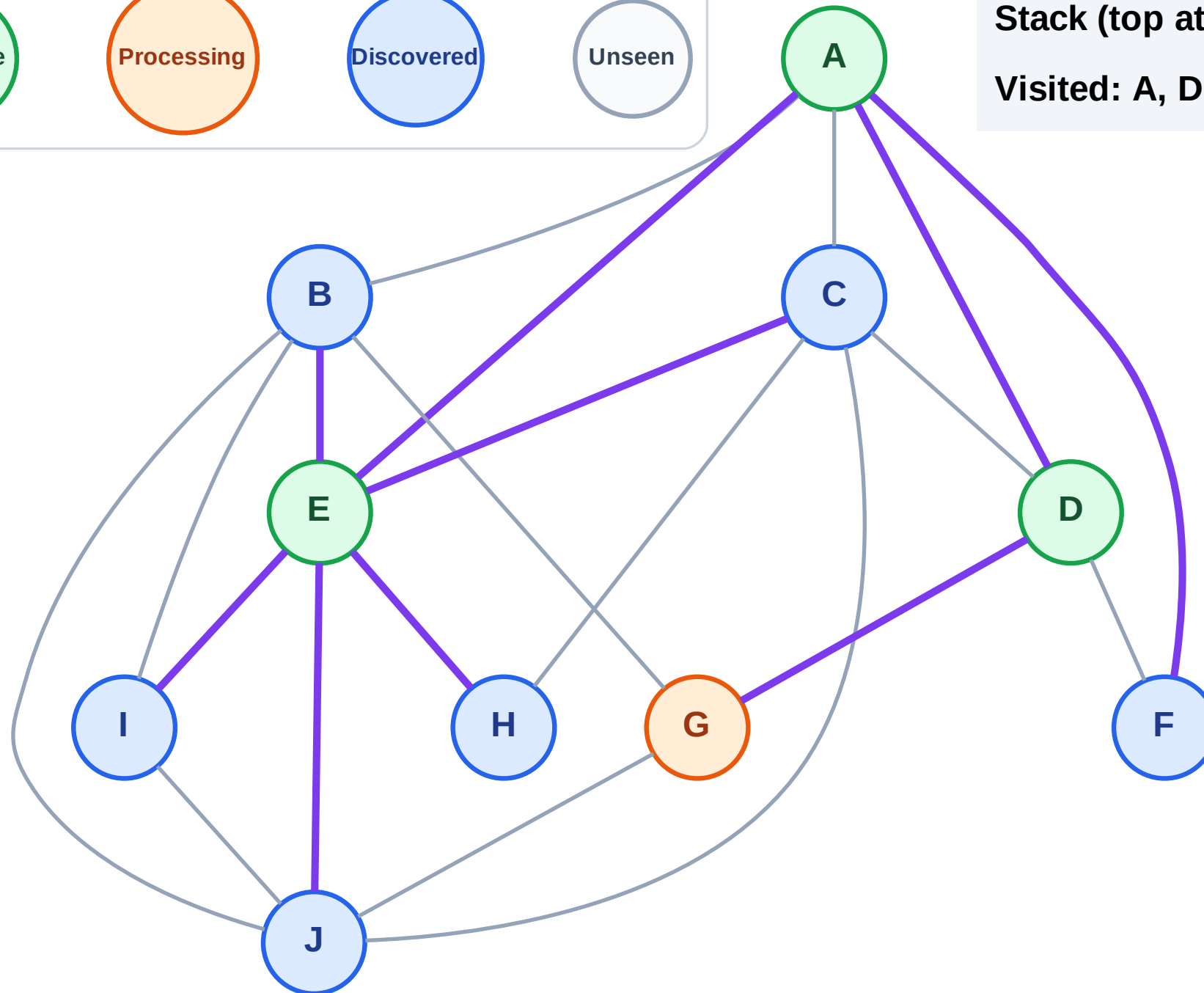
Step 8: Process node G

Legend



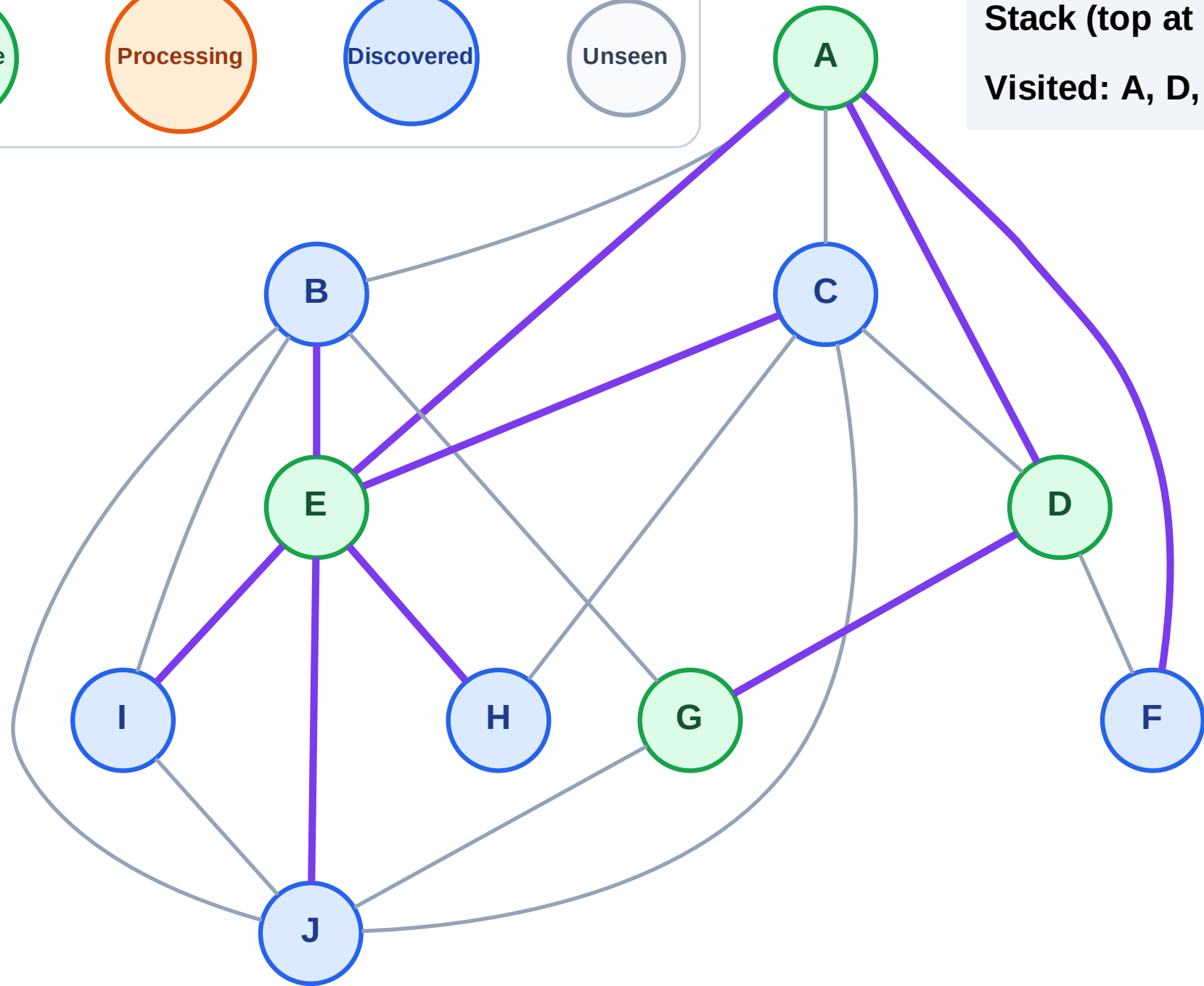
Stack (top at right): J | I | H | C | B | F

Visited: A, D, E



Step 9: No new neighbors from G

Step 9: No new neighbors from G

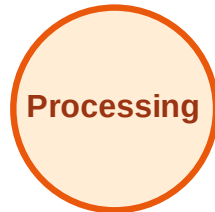


Visited: A, D, E, G

DFS Traversal

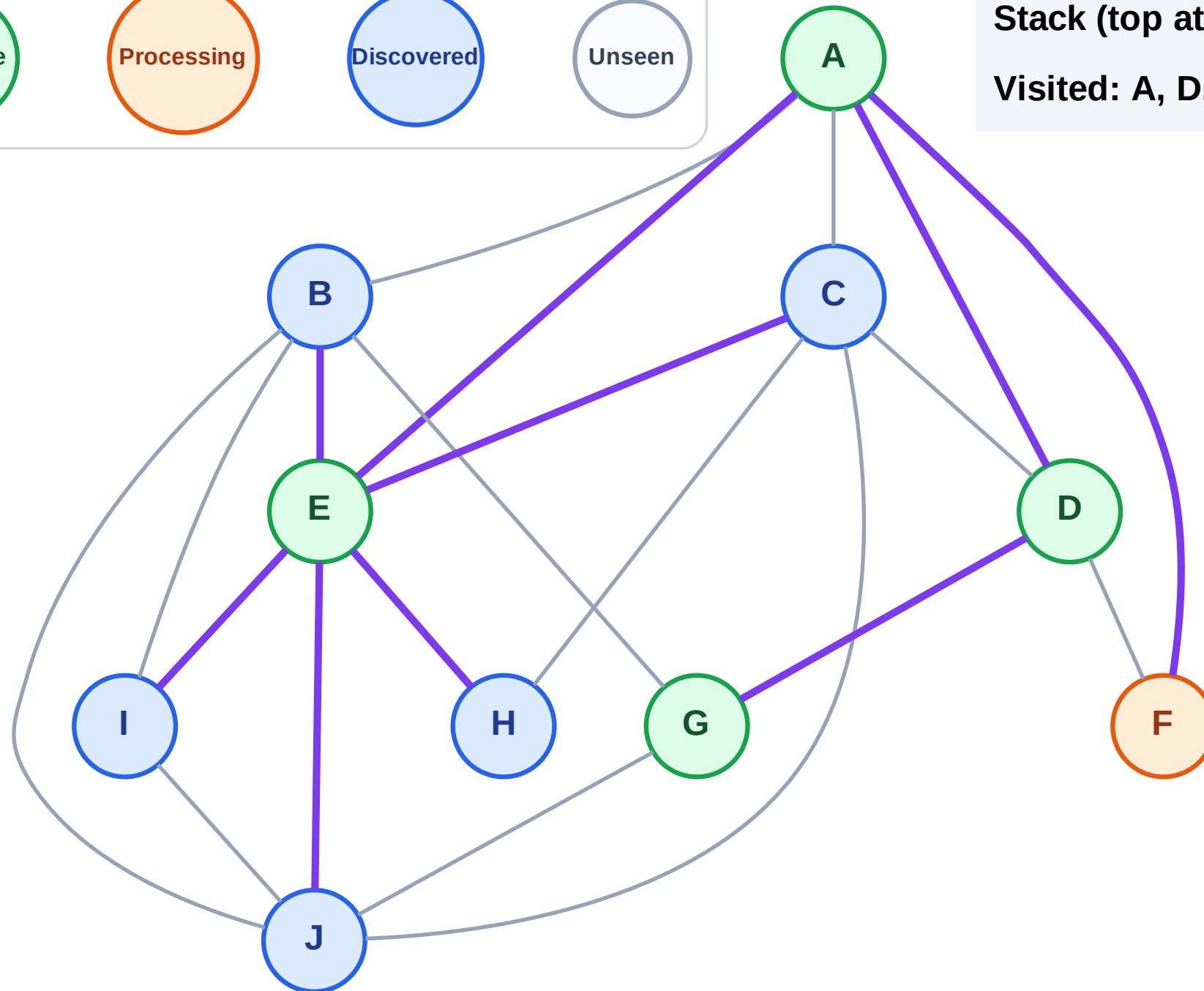
Step 10: Process node F

Legend



Stack (top at right): J | I | H | C | B

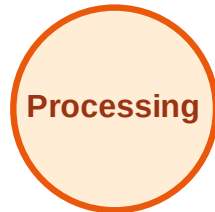
Visited: A, D, E, G



DFS Traversal

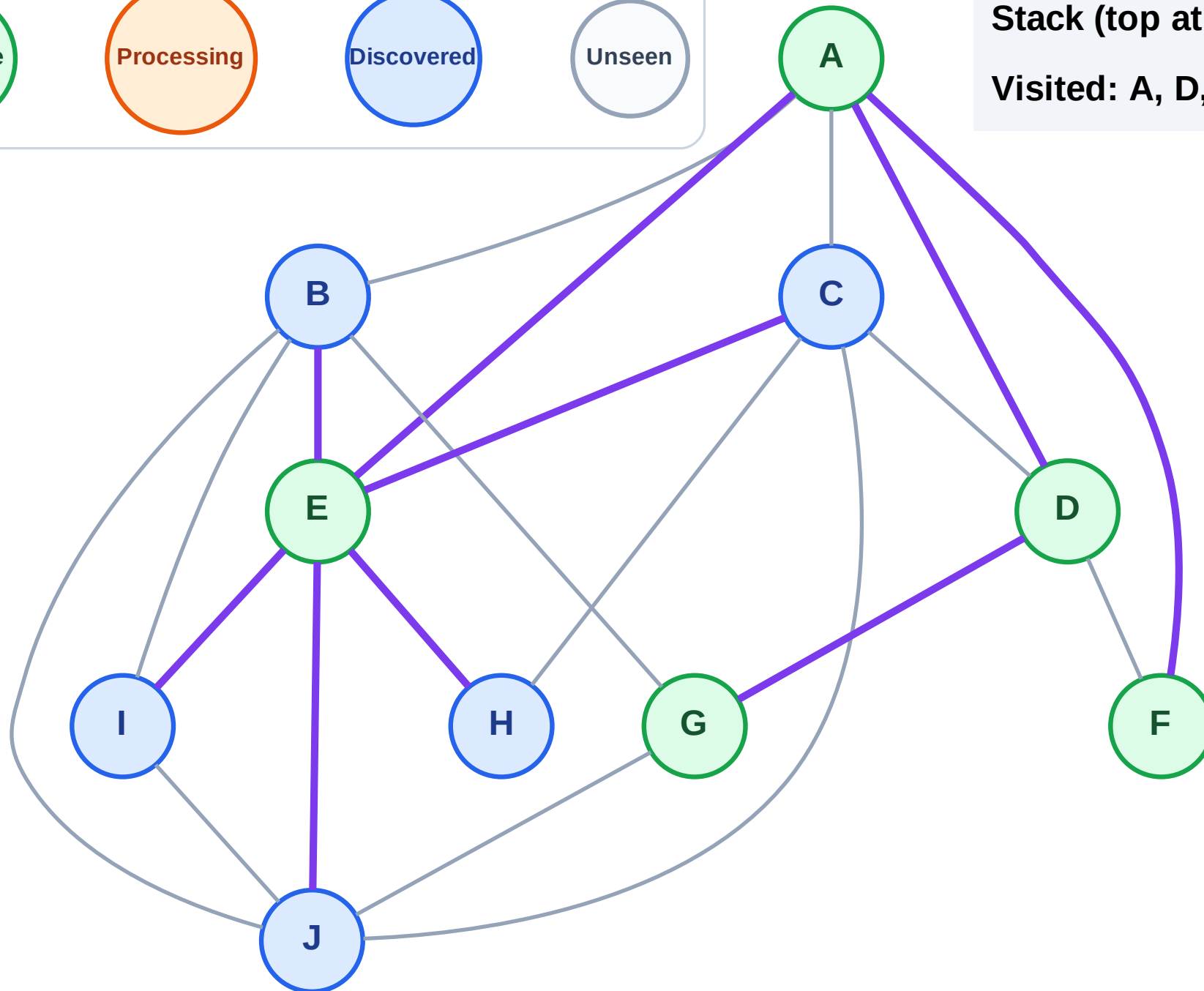
Step 11: No new neighbors from F

Legend



Stack (top at right): J | I | H | C | B

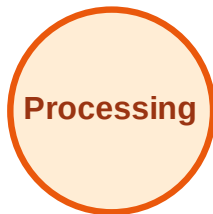
Visited: A, D, E, F, G



DFS Traversal

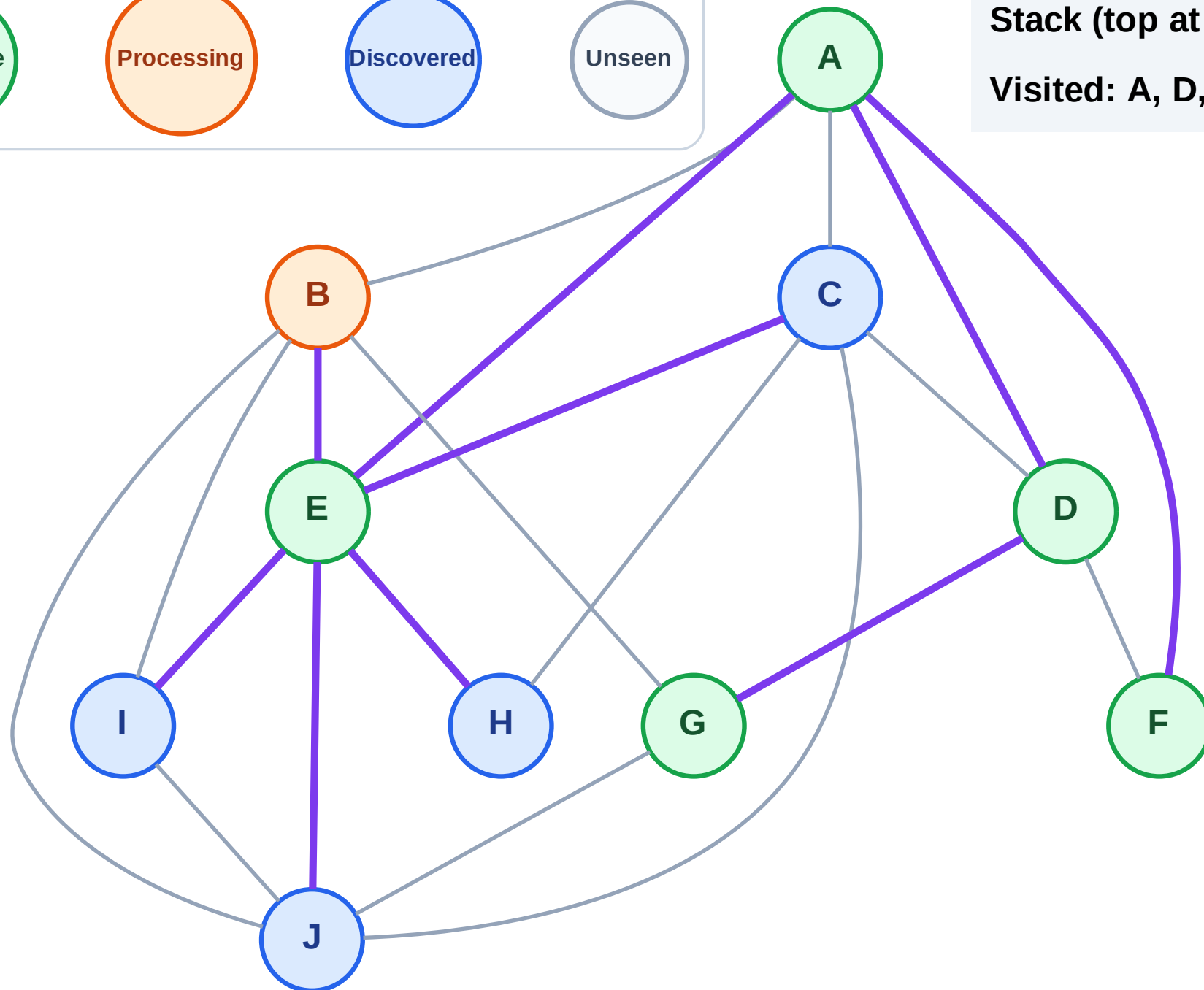
Step 12: Process node B

Legend



Stack (top at right): J | I | H | C

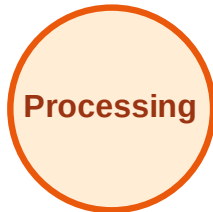
Visited: A, D, E, F, G



DFS Traversal

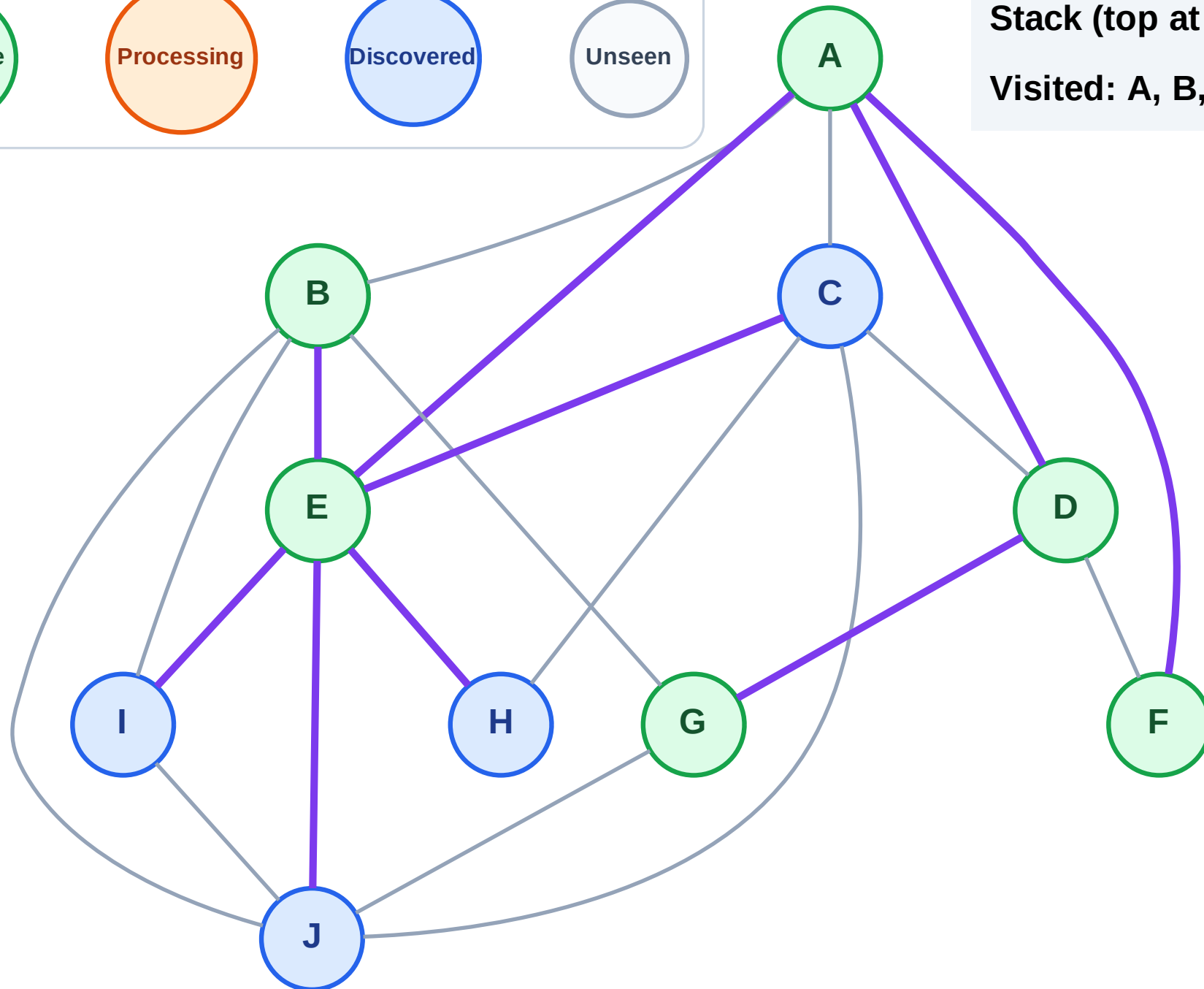
Step 13: No new neighbors from B

Legend



Stack (top at right): J | I | H | C

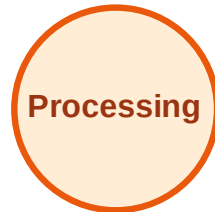
Visited: A, B, D, E, F, G



DFS Traversal

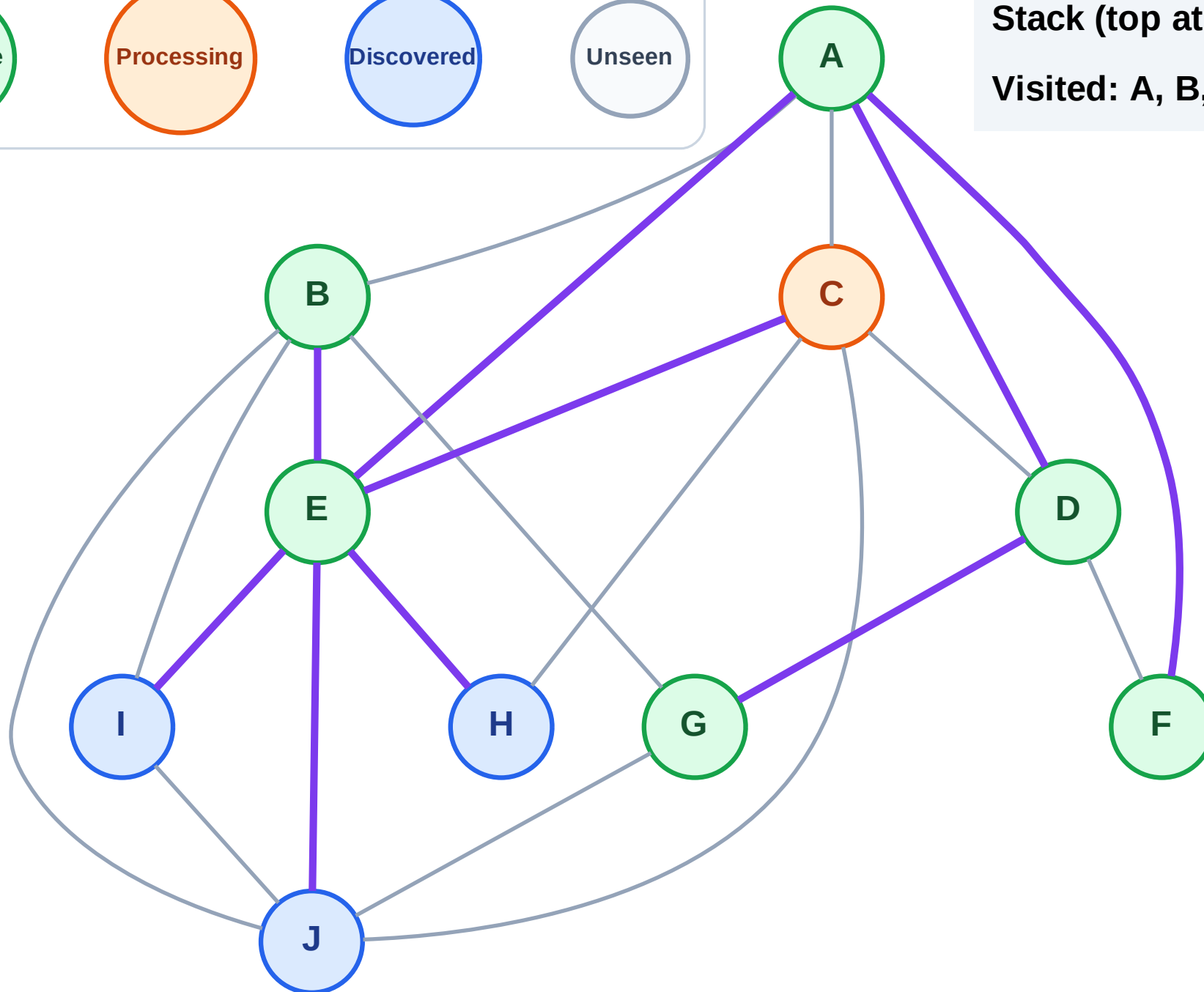
Step 14: Process node C

Legend



Stack (top at right): J | I | H

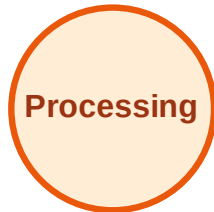
Visited: A, B, D, E, F, G



DFS Traversal

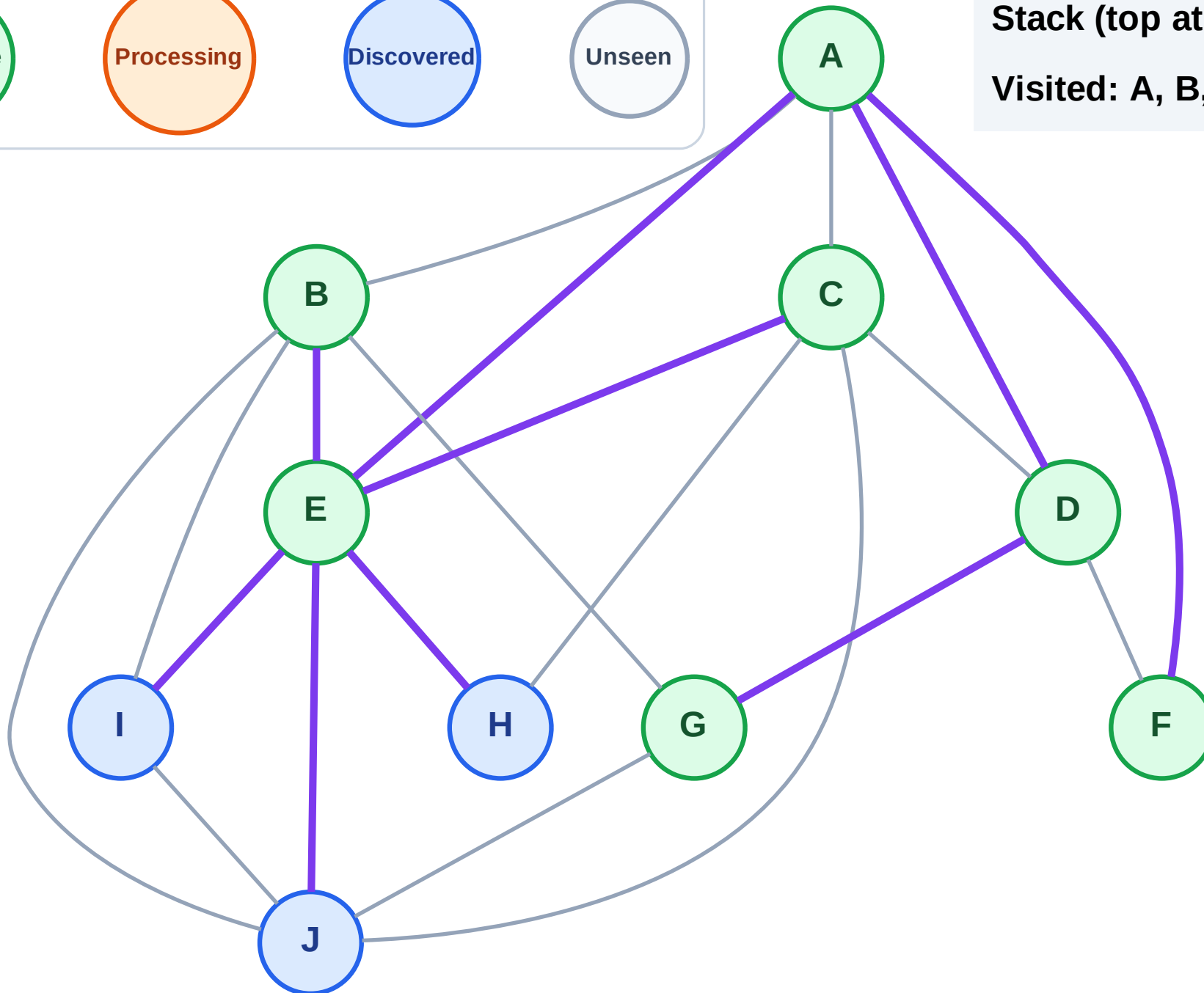
Step 15: No new neighbors from C

Legend



Stack (top at right): J | I | H

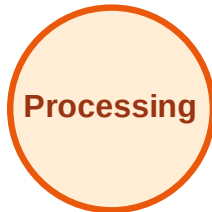
Visited: A, B, C, D, E, F, G



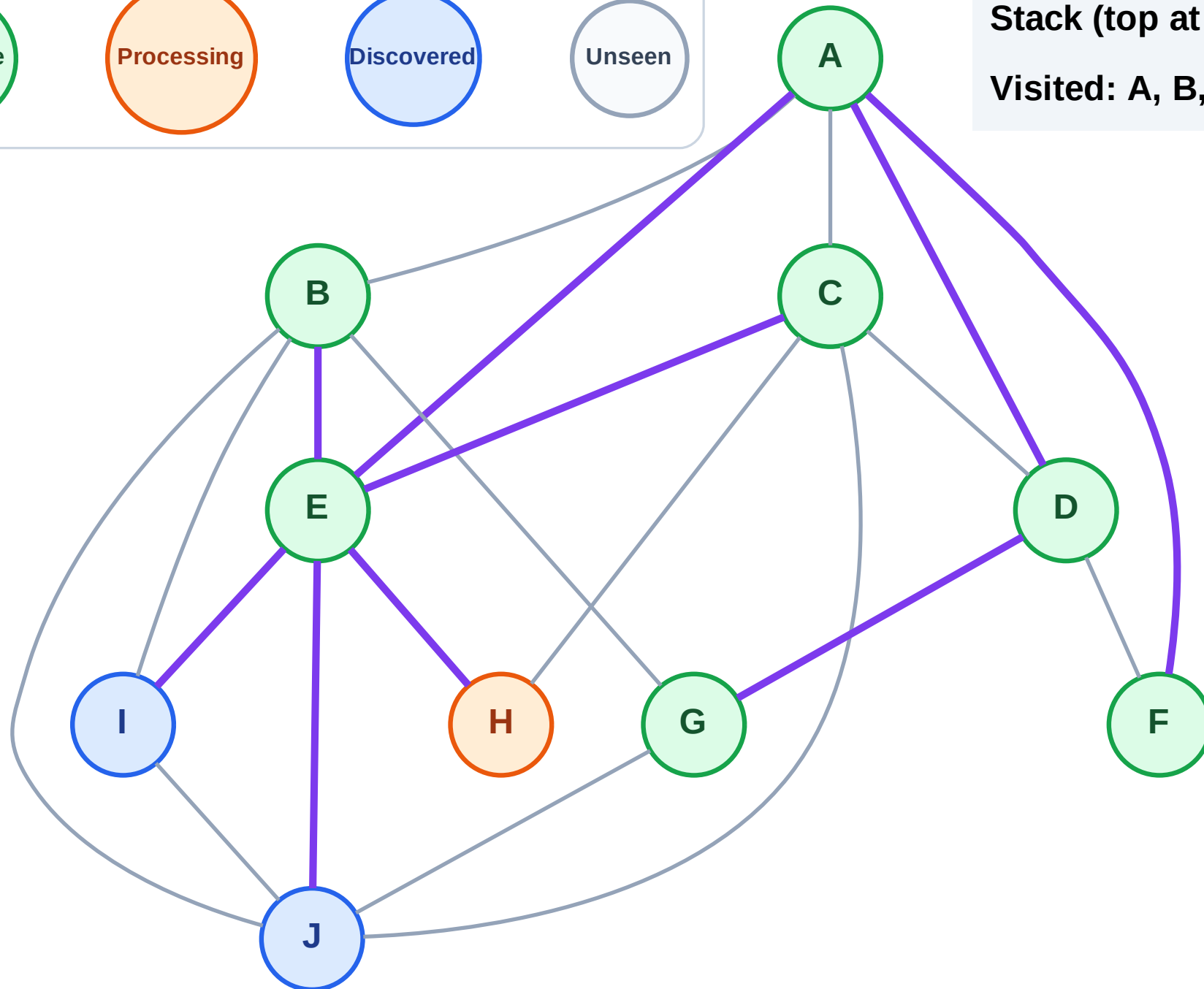
DFS Traversal

Step 16: Process node H

Legend



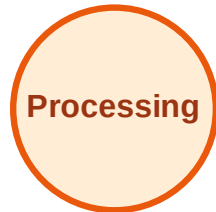
Stack (top at right): J | I
Visited: A, B, C, D, E, F, G



DFS Traversal

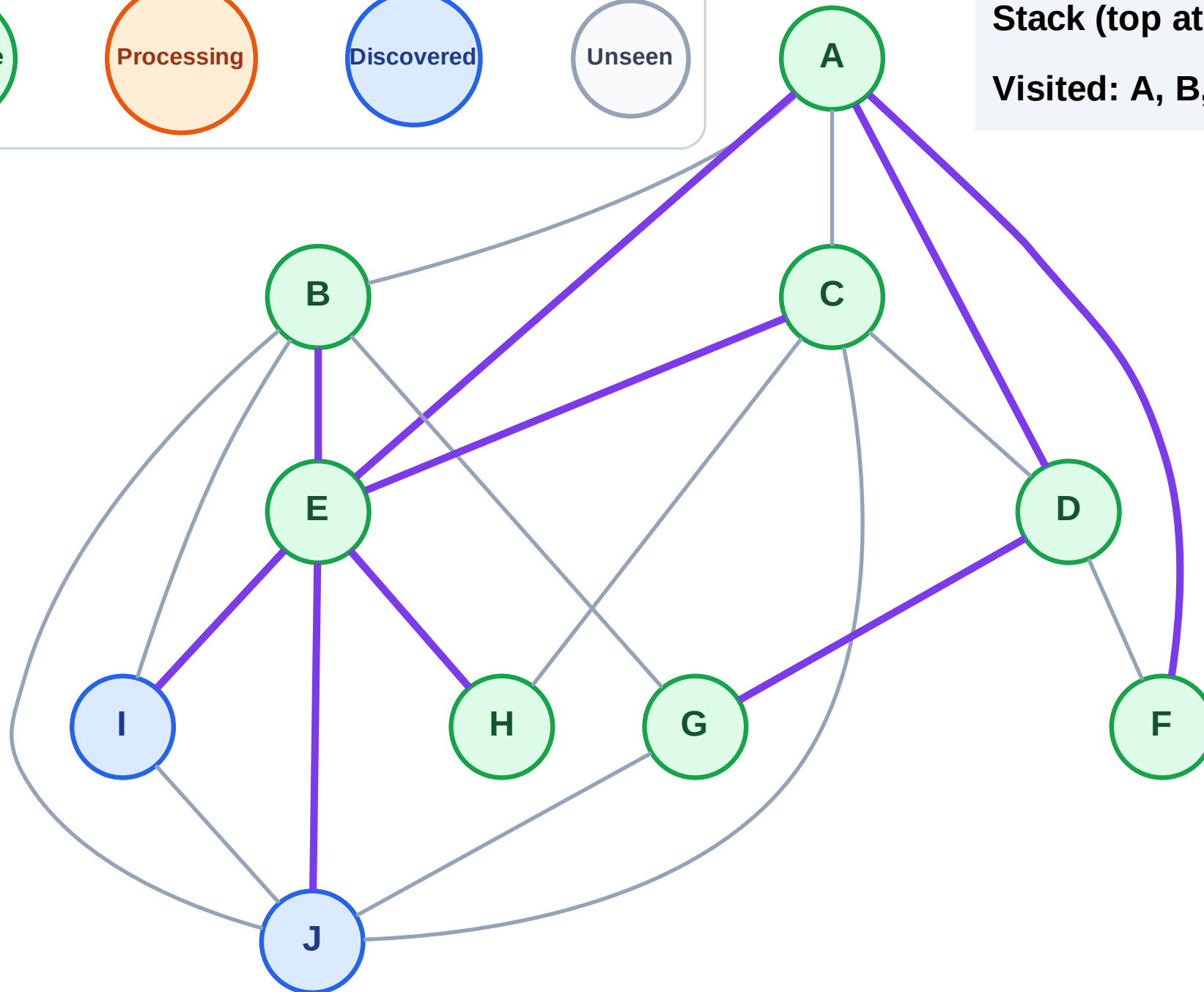
Step 17: No new neighbors from H

Legend



Stack (top at right): J | I

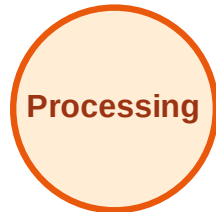
Visited: A, B, C, D, E, F, G, H



DFS Traversal

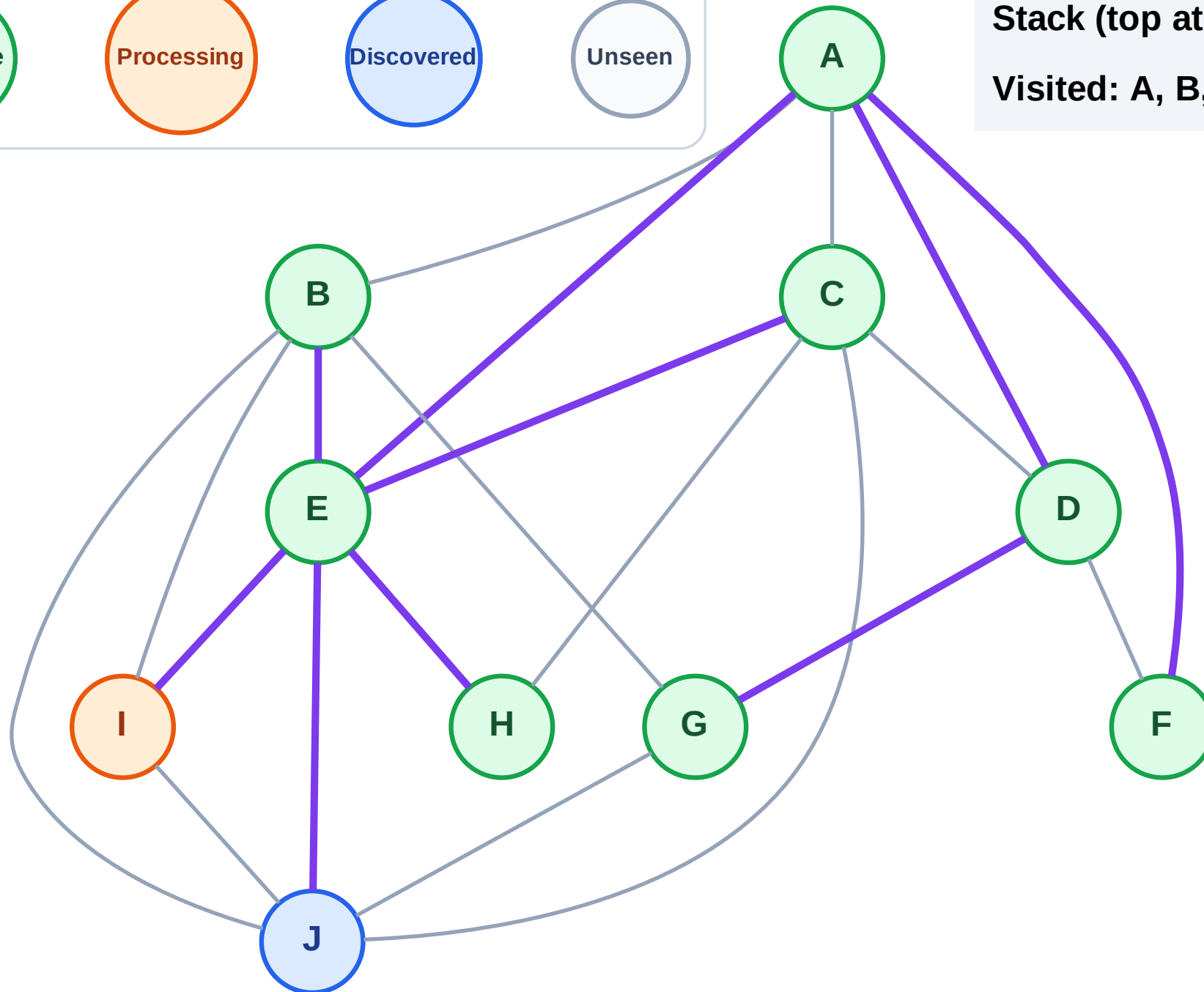
Step 18: Process node I

Legend



Stack (top at right): J

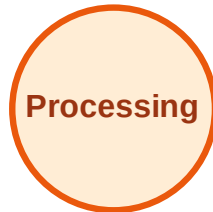
Visited: A, B, C, D, E, F, G, H



DFS Traversal

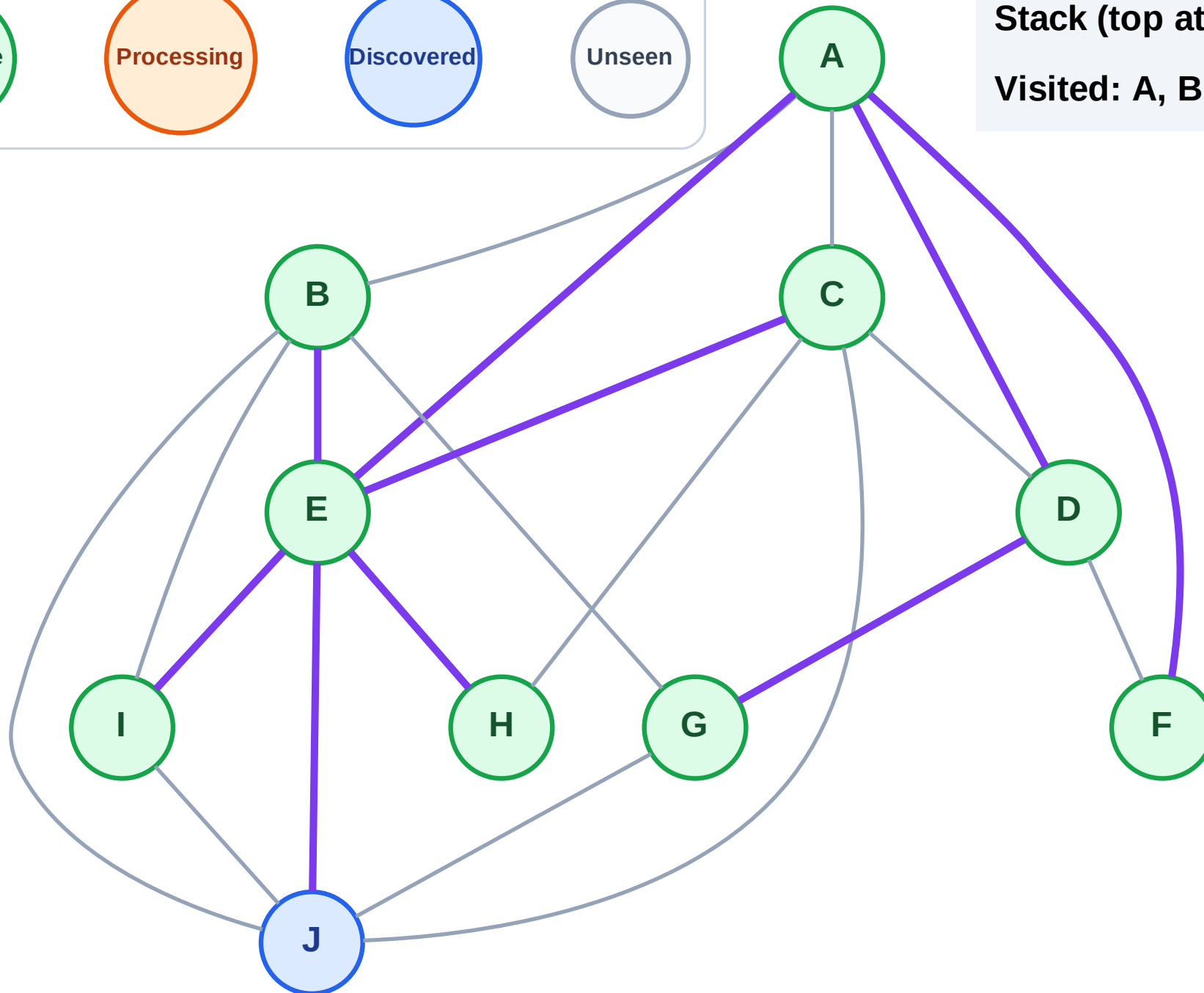
Step 19: No new neighbors from I

Legend



Stack (top at right): J

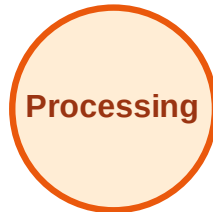
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

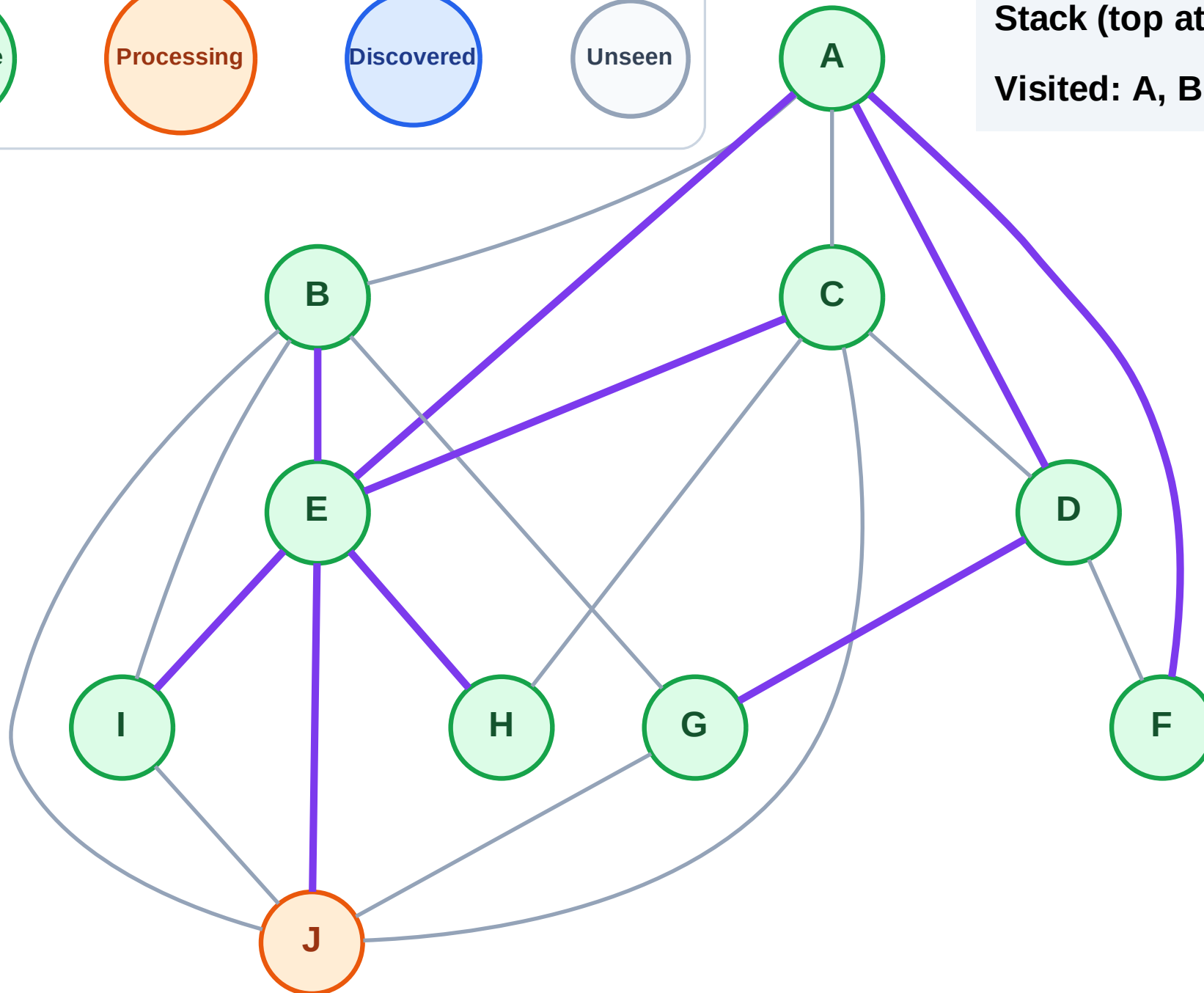
Step 20: Process node J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

